

Ex vivo Expansion Potential of Murine Hematopoietic Stem Cells: A Rare Property Only Partially Predicted by Phenotype

Reviewed Preprint

Revised by authors after peer review.

[About eLife's process](#)

Reviewed preprint version 2

January 24, 2024 (this version)

Reviewed preprint version 1

November 7, 2023

Posted to preprint server

September 9, 2023

Sent for peer review

September 8, 2023

Qinyu Zhang, Rasmus Olofzon, Anna Konturek-Ciesla, Ouyang Yuan, David Bryder 

Division of Molecular Hematology, Department of Laboratory Medicine, Lund Stem Cell Center, Faculty of Medical, Lund University, 221 84 Lund, Sweden

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

The scarcity of hematopoietic stem cells (HSCs) restricts their use in both clinical settings and experimental research. Here, we examined a recently developed method for expanding rigorously purified murine HSCs *ex vivo*. After three weeks of culture, only 0.1% of cells exhibited the input HSC phenotype, but these accounted for almost all functional long-term HSC activity. Input HSCs displayed varying potential for *ex vivo* self-renewal, with alternative outcomes revealed by single cell multimodal RNA- and ATAC-seq profiling. While most HSC progeny offered only transient *in vivo* reconstitution, these cells efficiently rescued mice from lethal myeloablation. The amplification of functional HSC activity allowed for long-term multilineage engraftment in unconditioned hosts that associated with a return of HSCs to quiescence. Thereby, our findings identify several key considerations for *ex vivo* HSC expansion, with major implications also for assessment of normal HSC activity.

Key point: *Ex vivo* self-renewal is an intrinsic property of rare candidate HSCs, with implications for assessments of HSC activity by transplantation.

eLife assessment

This study presents a **valuable** dissection on how functional HSCs are expanded in PVA cultures. The functional and multi-omic analyses provided are **convincing**, although the additional data and their analysis provided during revision could have been included in the test to assist readers and to strengthen the published manuscript. Nevertheless, the present work will be of value for stem cell biologists interested in HSC regulation.

Introduction

Hematopoietic stem cells (HSCs) are the functional units in bone marrow transplantation (BMT) and gene therapy approaches via their ability for multilineage differentiation and long-term self-renewal^{1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100}. However, restrictions in HSC numbers, the requirement of harsh host conditioning

and challenges with how to genetically manipulate HSCs with retained function are barriers for BMT/gene therapy². Due to the lack of reliable models for accurate assessment of human HSCs, mice continue to be the preferred experimental model for studying HSC biology. However, studying murine HSC biology also poses challenges due to the scarcity of HSCs.

Recently, culture conditions supporting efficient maintenance and expansion of murine HSCs were reported³. However, the expansion of functional *in vivo* HSC activity over a 28-day period (estimated to 236-899-fold) was not on par with the total cell expansion ($>10^4$ -fold), demonstrating that large amounts of differentiated progeny were generated along with self-renewing HSC divisions³. Moreover, candidate HSCs (cHSCs) by phenotype represented only a minor portion of the cells at the culture end-point³, although assessment of HSC activity based on this might be obscured by previous observations that HSCs change some phenotypes in culture⁴. While *in vitro* differentiation complicates assessments of HSC expansion, it might also be beneficial. For example, while HSCs can support lifelong hematopoiesis after bone marrow transplantation (BMT), they have limited capacity to rescue the host from lethal conditioning⁵. To address this limitation, there is a requirement for progenitor cells that can efficiently supply the host with mature blood cells, such as erythrocytes and platelets^{6,7}.

When transplanted, *in vivo* HSC activity is typically determined using both quantitative and qualitative parameters⁸. Most commonly, this is achieved by comparing to a known amount of competitor cells⁹. However, although the competitive repopulation assay (CRA) represents a mainstay in experimental HSC biology, it associates with shortcomings often neglected. First, while clinical BMT aims at transplanting large numbers of HSCs^{10,11}, experimental murine BMT typically use limited HSC grafts, which can strongly influence on HSC function¹². Secondly, the CRA is from a quantitative perspective rather blunt, with sigmoidal rather than linear characteristics⁹. This limits the boundaries of the effects that can be measured. Thirdly, the CRA assesses HSC activity based on overall hematopoietic chimerism, which might reflect poorly on the actual HSC activity given the vastly different turnover rates of different blood cells lineages¹³. At the same time, alternative assays such as limit dilution assays or single-cell transplantation assays are cumbersome, expensive and associated with ethical concerns. Newer developments in DNA barcoding technology have overcome many of these issues¹⁴.

Another concern with BMT is the use of aggressive host conditioning to permit engraftment. While eradicating malignant cells is an important part of treatment regimens for leukemic patients, the fates of HSCs in conditioned hosts versus native hematopoietic contexts are dramatically different¹⁵. But transplantation into unconditioned recipients is inefficient and requires large amounts of transplanted HSCs to saturate available niches^{16,17}. To achieve even limited HSC chimerism in unconditioned hosts can have strong long-term therapeutic effects¹⁸ and many experimental approaches might benefit from avoiding host conditioning.

Here, we provide molecular and functional details on HSC differentiation and self-renewal following culture and define a set of parameters critical for *in vivo* assessment of HSC activity. Importantly, we find that HSC heterogeneity was a key influencing factor for HSC expansion even among rigorously purified cHSCs. This determinism has broad implications for experiments involving *in vitro* manipulation of HSCs, but also when assessing normal HSC function by BMT.

Results

***In vivo* HSC activity is restricted to cells expressing high levels of EPCR**

We first assessed the *in vivo* reconstituting activity of cHSCs with different phenotypes. The endothelial protein C receptor (EPCR or PROCR/CD201) is highly expressed on HSCs¹⁹, while CD41 (ITGA2B) was suggested to be functionally relevant for HSC maintenance and homeostasis²⁰. Based on expression of EPCR and CD41, we isolated 4 subfractions within BM SLAM LSK cells (Lin⁻ Sca1⁺ cKit⁺ CD150⁺ CD48^{-/low}) (Fig 1A) and competitively transplanted them at equal numbers into lethally irradiated recipients (Fig 1B). The reconstitution from the three EPCR^{high} fractions was restricted to the PB myeloid lineages 4-week post-transplantation (Fig 1C), and most of these recipients displayed robust long-term (beyond 16 weeks) multilineage reconstitution with equivalent cHSC chimerism in the BM (Fig 1C-D). EPCR^{high}CD41⁻ cells produced slightly more lymphoid cells than the EPCR^{high}CD41⁺ cells at the endpoint (Fig S1), but the cHSC chimerism in the BM was very similar to other tested EPCR^{high} fractions (Fig 1D). In contrast to Gekas et al.²⁰, we failed to observe that staining with the CD41 antibody (MWreg30 clone) compromises HSC engraftment. Transplantation of EPCR negative cells failed to associate with either long-term PB or BM cHSC chimerism, demonstrating lack of HSC activity within this fraction (Fig 1C-D).

Next, single EPCR⁺ SLAM LSKs were index sorted (Fig S1B) and expanded *ex vivo* in an F12-Polyvinyl alcohol (PVA) based culture media²¹. At an early stage (8 days, Fig S1C), cultures initiated with SLAM LSK cells with higher EPCR expression proliferated less compared to those with lower EPCR expression. However, after longer expansion (25 days, Fig S1C), the EPCR higher cells generated on average a larger amount of progeny. To evaluate the HSC activity following *ex vivo* expansion and to correlate this directly to EPCR expression levels, the expanded cells from index sorted cultures of cHSCs were collected and transplanted into individual lethally irradiated mice (Fig S1D). While freshly and stringently isolated HSCs normally fail to rescue mice from lethal irradiation⁵, we here entertained that culturing of HSCs in addition to inducing self-renewal, would also generate progenitors that could radioprotect the hosts. If so, survival could be used as a proxy for combined hematopoietic stem and progenitor cell (HSPC) activity. Indeed, mice receiving progeny from SLAM LSK EPCR higher cells were more efficiently radioprotected (Fig 1E). Thereafter, we initiated cultures with EPCR^{high} or EPCR^{lower} SLAM LSK cells fractions (Fig S1E). Mice transplanted with cells expanded from EPCR^{high} cells radioprotected 5/5 recipients, while only two out of 5 animals receiving the progeny of EPCR^{low} cells survived long-term. As expected, these surviving mice displayed exclusive test-cell derived myelopoiesis long-term after transplantation (Fig 1F).

Taken together, these experiments established that HSC activity was restricted to EPCR^{high} HSCs, and such cells could effectively generate progeny *in vitro* that rescue mice from lethal irradiation. Therefore, we operationally defined our input cHSCs by their EPCR^{high} SLAM phenotype.

Phenotypic heterogeneity from *ex vivo* expanded cHSCs

The PVA-dependent cell culture system was reported to efficiently support HSC activity *ex vivo* over several weeks³. However, the functional HSC frequency did not appear on par with the total amount of cells generated by the end of the culture period, suggesting the generation also of many differentiated cells³. To detail this, we expanded aliquots of cHSCs *ex vivo* for 14-21 days, followed by multicolor phenotyping. From 50 cHSCs, an average of 0.5 million (14 days) and 13.6 million cells (21 days) were generated (Fig S2A) and which associated with a phenotypic heterogeneity that increased over time (Fig S2B). With the assumption that HSCs might retain their cell surface phenotype in cultures, we quantified EPCR^{high} SLAM cells and observed a decrease in

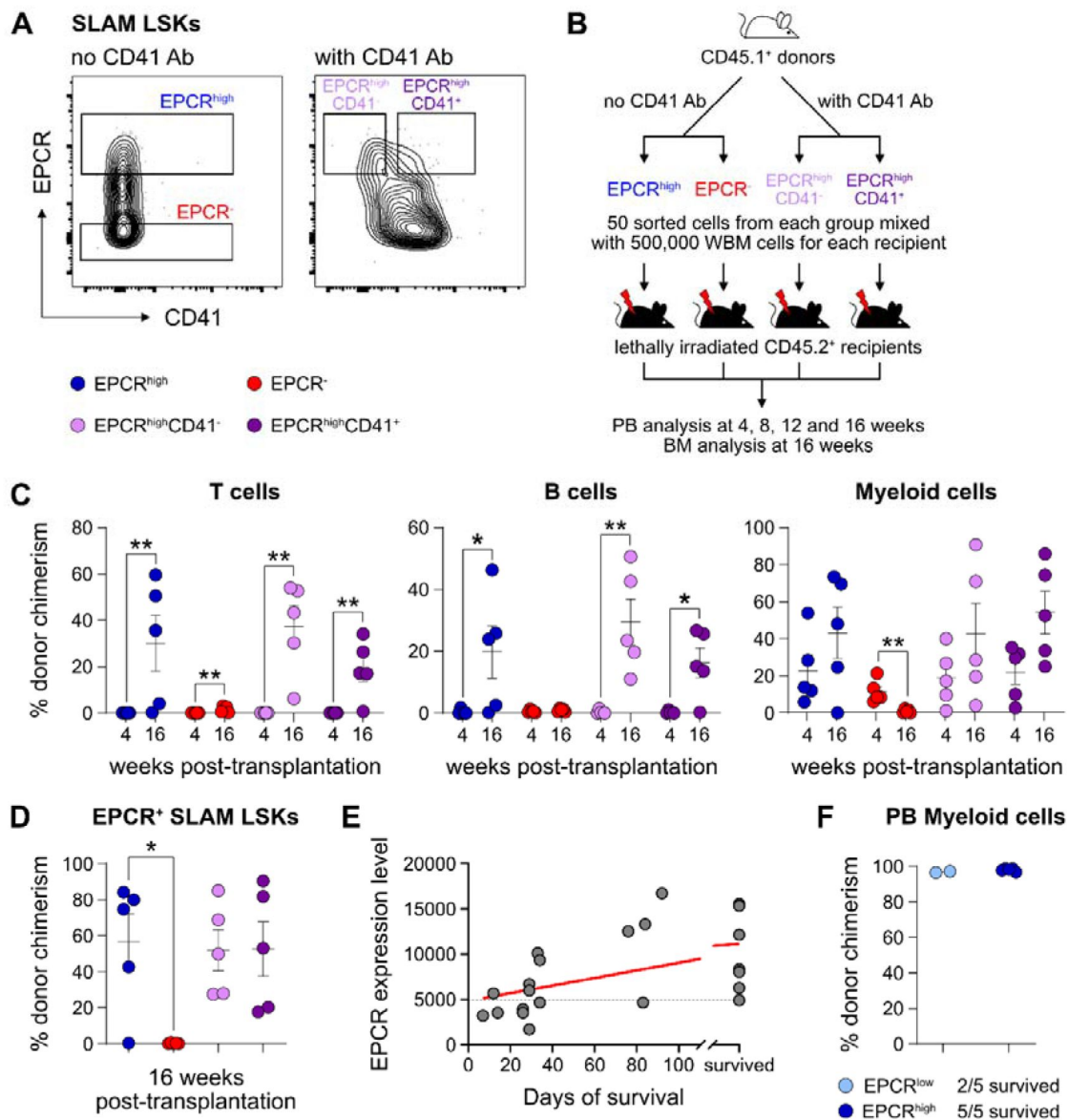


Figure 1.

EPCR expression within the BM SLAM LSK fraction is a high-confidence predictor of transplantation-associated HSC activity.

(A) Expression patterns of EPCR and CD41 within the BM LSK SLAM fraction. Gates depict the assessed cell fractions.

(B) Strategy used to assess the correlation of EPCR and CD41 expression to the *in vivo* HSC activity.

(C) Test-cell derived chimerism in PB 4 and 16 weeks post-transplantation.

(D) Test-cell derived chimerism in BM EPCR⁺ SLAM LSKs 16 weeks post-transplantation.

(E) Correlation between duration of radioprotection and EPCR expression levels. BM LSK SLAM cells were co-stained with EPCR and index-sorted at one cell per well, cultured for 21 days, and the content of each well transplanted to individual recipients ($n = 22$). Correlation to mortality of individual mice was made by assessing which well was transplanted into which mouse and coupling this to the index-sorting information. The grey dash line indicated the separation of EPCR higher (>4900) or lower expression (<4900) cHSCs.

(F) Donor contribution in PB myeloid cells. Mice were transplanted with *ex vivo* expanded cells from either 50 SLAM LSK EPCR^{low} ($n = 5$) or 50 SLAM LSK EPCR^{high} cells ($n = 5$) and assessments made 16 weeks after transplantation.

All data points depict values in individual recipients. Error bars denote SEM. The asterisks indicate significant differences. *, $p < 0.05$; **, $p < 0.01$. In E) a regression line was generated based on an end-point survival of 150 days (the time at which the experiment was terminated).

their frequency over time, although this was numerically counteracted by the overall proliferation in cultures (**Fig 2A** [↗](#)). Thus, quantification of cHSCs translated into an average expansion of 54-fold (26- to 85-fold) and 291-fold (86- to 1,273-fold) at 14 and 21 days, respectively (**Fig 2A** [↗](#)).

To gain a deeper understanding of the cells generated in our *ex vivo* cultures, we collected cells from the entire culture expanded from 500 cHSCs and isolated nuclei for single cell multiome sequencing using a commercial platform from 10x Genomics. This approach combines single-cell RNA sequencing and single-cell ATAC sequencing, allowing for integrated analysis of these modalities.

Based on the transcriptomic signatures and distal motif identities, we were able to identify not only early hematopoietic stem and progenitor cells (HSPCs), but also many differentiated myeloid cells in the expansion cultures (**Fig 2B** [↗](#)), however with little evidence for lymphoid differentiation. Assessment of a condensed HSC signatures (**Fig S2C**) confirmed that the cHSCs were primarily located to cluster 5 (**Fig 2D** [↗](#)), with trajectory analyses suggesting multiple differentiation pathways from early HSPCs to differentiated myeloid progeny (**Fig 2C** [↗](#)).

Since the EPCR expression also condensed to cluster 5 (**Fig S2D**), we further investigated whether the expression of EPCR marks cHSCs *ex vivo* by collecting *ex vivo* expanded EPCR⁺ cells for single-cell multiome sequencing. These cells displayed less heterogeneity than what we observed for the total culture (**Fig 2B** [↗](#)), which included a more even distribution of the condensed HSC signature (**Fig 2E** [↗](#)) as well as the expression of EPCR (**Fig S2E**). Here, the most pronounced separator for the interrogated cells attributed to the cell cycle position (**Fig 2F** [↗](#)).

To explore if other markers could provide additional phenotypic information for genuine HSCs in cultures, we applied an Fgd5 reporter mouse model previously shown to selectively mark HSCs *in situ*²²[↗](#). While the Fgd5 reporter only marked a subset of cells in culture (**Fig S2F**), we noted heterogeneous expression of CD48 within the Fgd5 positive fraction (**Fig S2F-G**) and that we failed to clearly demarcate in our single-cell multimodal data.

Collectively, these results demonstrated robust expansion of cHSCs using the PVA-based culture system, albeit with additional parallel generation of a large number of more differentiated cells. While the identities of most of the cells in cultures could be clearly deduced by their combined gene expression and chromatin accessibility profiles, this approach failed to inform on the heterogeneous expression pattern of CD48 on cHSCs (**Fig S2**).

Functional HSC activity associates to a minor LSK

SLAM EPCR^{high} fraction within *ex vivo* cultures

Given our inability to distinctly identify cHSCs by their molecular profiles, we next explored the possibility to prospectively isolate HSCs from the cHSC cultures and instead rely on their functional ability to long-term repopulate lethally irradiated hosts for readout. cHSCs were isolated and cultured *ex vivo* for 21 days and the expanded cells were separated into two equal portions: one portion was kept unfractionated, while three subpopulations of CD150⁺LSKs: EPCR^{high}CD48^{low}, EPCR^{high}CD48⁺ and EPCR⁺ were sorted from the other portion. The fractions were competitively transplanted into lethally irradiated mice, such that each recipient received fractions equivalent to the expansion from 50 cHSCs (EE50) along with 500,000 competitor WBM cells (**Fig 3A** [↗](#)).

Stable and very high long-term multilineage reconstitution was observed in all recipients of unfractionated cultured cells. By contrast, test-cell derived reconstitution was not recovered from EPCR⁺ cells (**Fig 3B** [↗](#) and **Fig S3**). EPCR^{high}CD48⁺ cells produced very high reconstitution levels short-term after transplantation, but the chimerism provided by these cells dropped considerably over time (**Fig 3B** [↗](#) and **Fig S3**). This differed strikingly from recipients receiving the minor

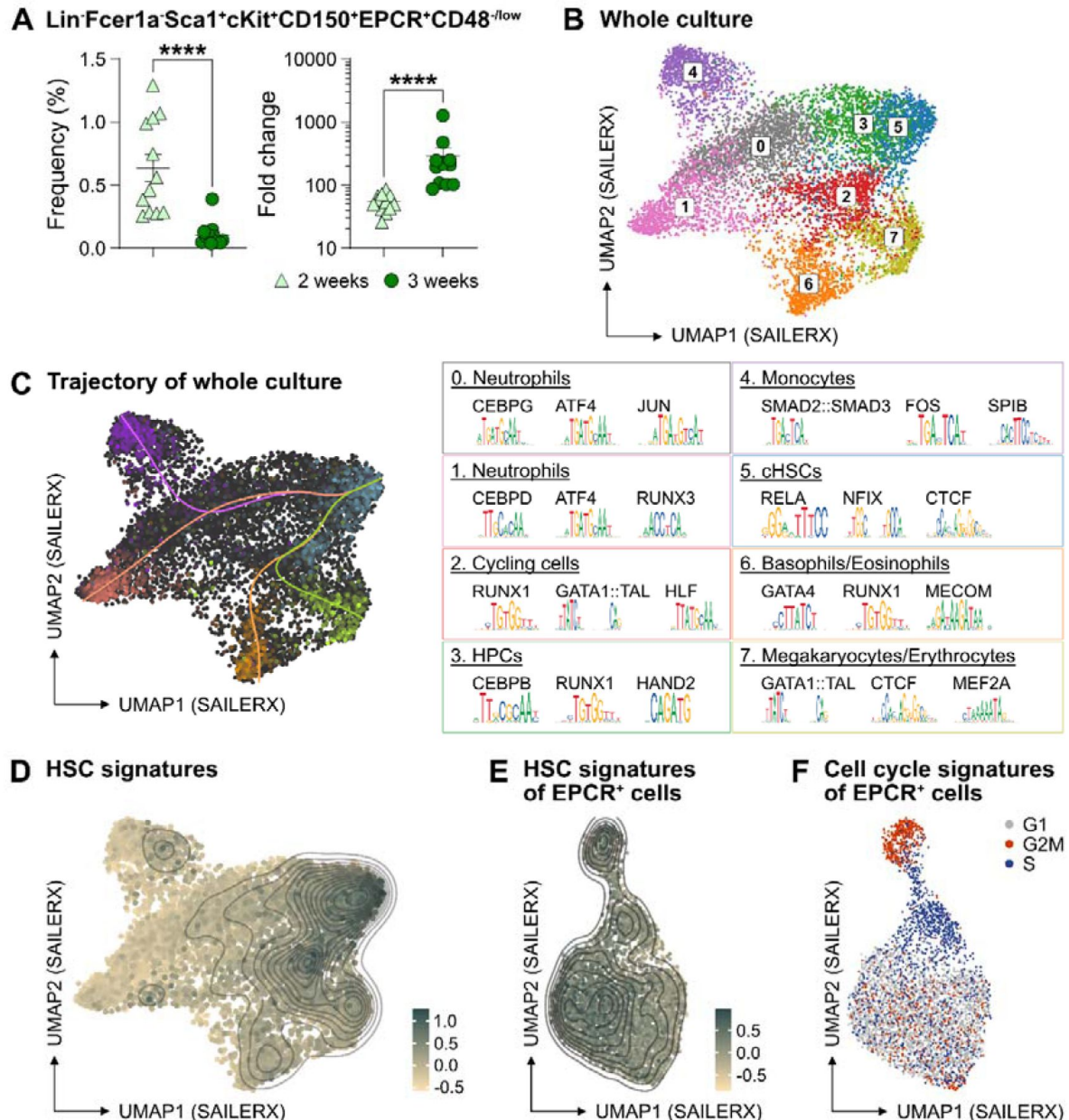


Figure 2.

The phenotypic properties of HSC expansion cultures.

(A) Frequency and fold change of phenotypic cHSCs (EPCR^{high} SLAM LSKs) in *ex vivo* cultures after 14 or 21 days of culture. Data points depict values from individual cultures initiated from 50 cHSCs. Error bars denote SEM. The asterisks indicate significant differences. ****, $p < 0.0001$.

(B) UMAP (based on SAILERX dimensionality reduction) of single-cell multiome profiling of cells expanded *ex vivo* for 21 days. Cell type annotations were derived using marker gene signatures and distal motif identities.

(C) Trajectory analysis of lineage differentiation for cells expanded *ex vivo* (left), with the top 3 scoring TF motifs of each cluster (right).

(D) Expression of HSC signature on whole culture.

(E) Expression of HSC signature of EPCR⁺ cells sorted from *ex vivo* cultures.

(F) Cell cycle phase classification of EPCR⁺ cells sorted from *ex vivo* cultures.

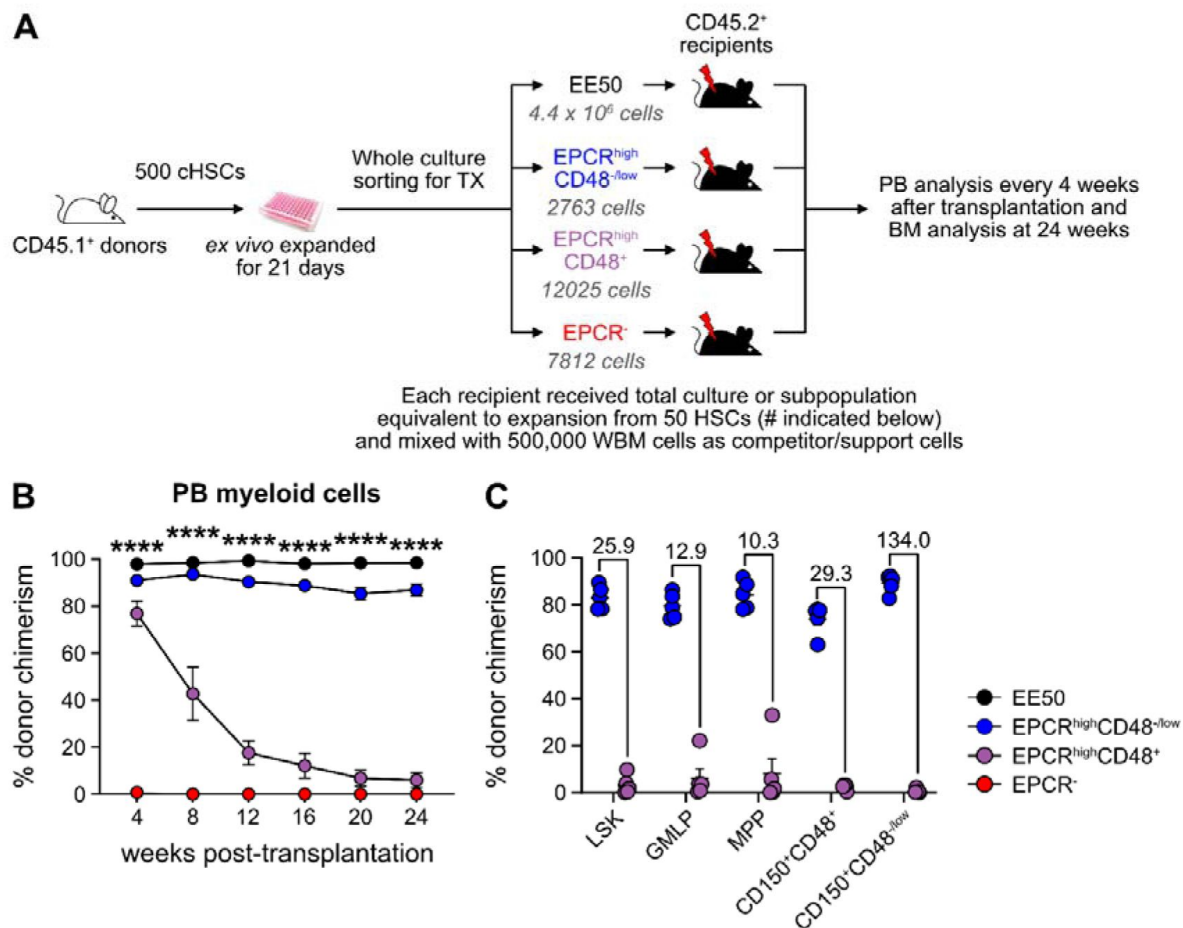


Figure 3.

HSC activity can be prospectively isolated from cHSC cultures and associates to a minor EPCR⁺ SLAM LSK fraction.

(A) Strategy to assess the *in vivo* HSC activity of subfractions from *ex vivo* cultures.

(B) Test-cell derived chimerism in PB myeloid lineages over 24-weeks post-transplantation. Data represent mean values ± SEM (n = 5 for each group). A one-way ANOVA test was applied and the asterisks indicate significant differences among the four groups. ****, p < 0.0001.

(C) Test-cell derived chimerism in BM progenitor subsets 24-weeks post-transplantation. Numbers indicate fold differences between the EPCR⁺ CD48^{low} and EPCR⁺ CD48⁺ fractions, and data points depict chimerism levels in individual recipients.

fraction of EPCR^{high}CD48^{low} cells, which presented with robust long-term multilineage reconstitution (**Fig 3B** and Fig S3). While almost no BM cHSC chimerism was observed from EPCR^{high}CD48⁺ cells, EPCR^{high}CD48^{low} cells produced very high chimerism in all evaluated progenitor fractions, including for cHSCs (**Fig 3C**).

These data established that functional *in vivo* HSC activity following culture is restricted to the minor fraction of SLAM LSK EPCR^{high} cells.

Quantification of HSC activity in *ex vivo* expansion cultures

Expanded cHSCs presented with a vastly higher reconstitution activity compared to freshly isolated cHSCs, and mostly fell out of range for accurate quantification (**Fig 3B**). We therefore assessed the impact of lowering the amount of input cHSCs and enhancing the amount of competitor cells. Aliquots of 10 cHSCs were cultured and expanded for 21 days. Cells from each culture (EE10) were next mixed with 2 or 20 million (2M or 20M) competitor WBM cells, followed by transplantation (**Fig 4A**, ‘individual’). In parallel, we also included a group in which EE50 cHSCs were mixed with 10 million WBM cells and split among 5 lethally irradiated recipients (**Fig 4A**, ‘pooled’). This allowed us to assess the variability among input cHSCs

Although robust reconstitution was observed in most recipients of ‘individual’ cultures (**Table 1**, **Fig 4B-C**, Fig S4A-B), the levels of test-cell derived cells varied extensively among these recipients. To quantify the HSC activity in each recipient, we calculated the repopulating units (RUs)⁹ for each lineage and recipient separately at the experimental endpoint (**Table 1**). Short-lived myeloid cells has been proposed as a better indicator of ongoing HSC activity²³, and in agreement with this, we observed a high correlation between the RU^{myeloid} and RUs for cHSCs in the BM (**Table 1**). A much more consistent HSC activity was observed in between recipients of ‘pooled’ cells (**Table 1**, **Fig 4B-C**). This confirmed that the differential chimerism in recipients of limited cHSC numbers relates to an unresolved cellular heterogeneity of input cHSCs. RUs per initial cHSC were calculated for freshly isolated BM cHSCs (the three EPCR^{high} groups from **Fig 1B**) and following 3-week culture (‘pooled’ group). This revealed 461- and 70-fold increases in PB myeloid and BM cHSC chimerism in recipients of *ex vivo* expanded cells, respectively. This is an estimate of the increase of functional HSC activity following *ex vivo* culture (Fig S4G).

We next assessed HSC clonality using a lentiviral barcoding approach. Five aliquots of 1,000 cHSCs were transduced with a barcode library²⁴ 48 hours after culture and then expanded for 19 days. Half of the expanded cells from each well were transplanted into one ‘parental’ recipient, respectively, while the other half was mixed with each other, with each ‘daughter’ recipient receiving 1/5 of the mixed cells (**Fig 4D-i**). As a proxy for the HSC activity, barcodes were extracted from BM test-derived myeloid cells 16-weeks post-transplantation. Following stringent filtering, we retrieved 223 unique barcodes in the ‘parental’ recipients, with a highly variable contribution (0.1 - 26.0%, **Fig 4E-G**). Representation of the same barcode in ‘parental’ and ‘daughter’ recipients demonstrates a shared clonal/HSC origin, where more robustly expanding HSCs should have a higher chance of reconstituting more ‘daughter’ recipients. Our data agreed well with this presumption (**Fig 4E**), and the most actively shared clones also associated with larger outputs in ‘daughter’ recipients (**Fig 4F**).

While our initial barcode experiments were designed to assess the *in vitro* expansion potential of cHSCs, a next set of experiments were executed to assess the clonal HSC activity following expansion. For this, cHSCs were *ex vivo* expanded for 13 days, provided with barcodes²⁵ overnight, and transplanted into 4 lethally irradiated recipients (**Fig 4D-ii**). 16-weeks post-transplantation, we recovered 573 unique barcodes. The median clone sizes of HSCs post-culture were significantly smaller than the pre-culture clones (**Fig 4G**), which was expected as more clones (573 vs 223) were evaluated in the ‘post-culture’ setting. However, the frequency of

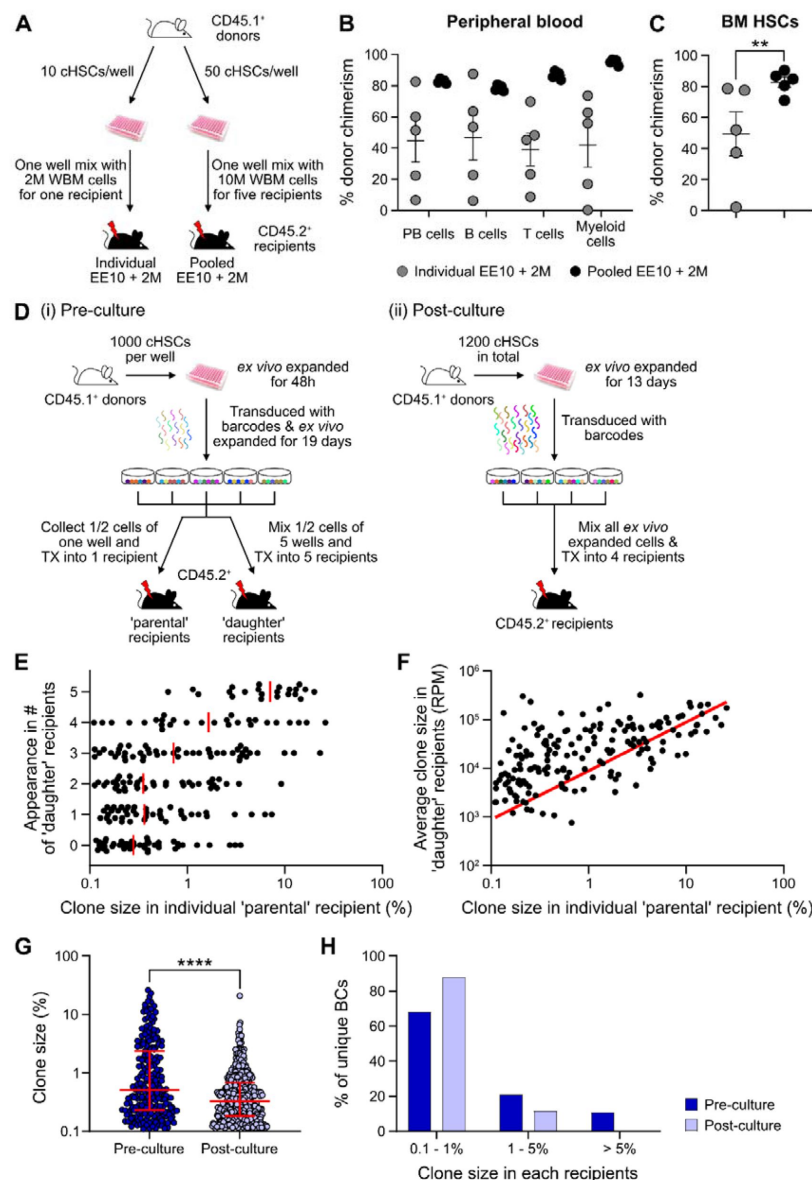


Figure 4

Quantification of repopulating activity from expanded cHSCs.

(A) Competitive transplantation strategies used to assess the repopulation of *ex vivo* expanded cHSCs.

(B) Test-cell derived PB reconstitution 16-weeks post-transplantation. Symbols denote individual mice and means $\pm$ SEM.

(C) BM cHSCs chimerism 16 weeks post transplantation. Symbols denote individual mice, and means $\pm$ SEM.

(D) Barcode approaches used to assess the clonal HSC contribution of *ex vivo* expanded HSCs before (i) or after (ii) *ex vivo* expansion.

(E) Clone sizes of unique barcodes in 'parental' recipients and their appearance in 'daughter' recipients, demonstrating extensive variation in expansion capacity among individual HSCs. Red lines indicate the median clone size in 'parental' recipients.

(F) Clone sizes of unique barcodes detected in BM myeloid cells of 'parental' recipients and their corresponding contribution in 'daughter' recipients. The red line denotes the correlation/linear regression.

(G) Clone sizes in 'parental' recipients transplanted with 'pre-culture' barcoded cells, or in recipients of 'post-culture' barcoded cells. Median clone sizes are shown with interquartile ranges.

(H) Frequency of barcodes and their clone sizes in recipients of pre- or post-cultured barcoded HSCs. BCs, barcodes.

All data points depict values in individual recipients or barcodes. Asterisks indicate significant differences. *, $p < 0.05$; ****, $p < 0.0001$.

<i>Recipients</i>		<i>PB</i>	<i>B cells</i>	<i>T cells</i>	<i>Myeloid cells</i>	<i>BM cHSCs</i>
Individual BM EE10 + 2M	#1	6	6	6	4	12
	#2	31	35	19	27	75
	#3	21	23	17	35	70
	#4	98	144	46	57	21
	#5	1	1	2	0.03	0.4
Pooled BM EE10 + 2M	#1	101	78	173	583	49
	#2	110	84	120	516	82
	#3	87	68	103	248	130
	#4	101	70	149	384	187
	#5	65	127	186	331	112

* 1 RU equals to the average reconstitution activity of 1×10^5 WBM cells.

Table 1.

RUs for each lineage in PB and for BM cHSCs of each recipient.

dominantly contributing clones was significantly less abundant (**Fig 4H** [↗](#)), demonstrating a more even contribution from individual clones in this setting. This is in line with the interpretation that expanded HSCs are functionally more equivalent than input cHSCs.

Finally, we assessed how the culture system supports the *in vivo* activity of fetal liver (FL) cHSCs. E14.5 FL-cHSCs were sorted using the same immuno-phenotype as BM cHSCs and cultured *ex vivo*. We observed an average 188-fold increase of cHSCs following 21-days culture (**Fig S4D**). We used the ‘pooled’ approach for functional evaluation and competed EE10 cells with either 2 or 20 million WBM cells (**Fig S4C**). While *ex vivo* expanded FL-cHSCs contributed considerably to multilineage reconstitution (**Fig S4E**), we observed compared to adult cHSCs 5.7- and 6.6-fold lower reconstitution from FL-cHSCs within PB myeloid cells and cHSCs, respectively (**Fig S4F-G**).

Taken together, these experiments demonstrated robust increases in HSC activity following culture of cHSCs, albeit slightly less for FL-cHSCs. However, clonal barcode assessments revealed substantial variation in *ex vivo* expansion potential of even stringently purified input cHSCs.

***Ex vivo* expanded cHSCs returns to a quiescent state following engraftment in unconditioned recipients**

Successful engraftment in an unconditioned setting requires large numbers of HSCs/HSPCs¹⁶ [↗](#), which reportedly can be accommodated by PVA-cultured HSCs³ [↗](#). To test this, we transplanted EE100 CD45.1⁺ cHSCs into lethally irradiated or unconditioned CD45.2⁺ recipients (**Fig S5A**). In lethally irradiated hosts, cultured cells reconstituted >90% of PB cells, including with robust contribution of donor cells to the myeloid lineages. By contrast, most of the unconditioned recipients lacked long-term reconstitution (**Fig S5B**). Given that CD45 mismatching might be a sufficient immunological barrier to prevent HSPC reconstitution²⁶ [↗](#), we instead transplanted expanded cells into unconditioned F1 CD45.1⁺/CD45.2⁺ hosts (**Fig 5A** [↗](#)). Clear long-term multilineage engraftment was observed in all these hosts (**Fig 5B** [↗](#)).

In the BM, cHSCs exist for the most part in a quiescent state, which contrasts the situation in cultures (**Fig S5C**). To investigate to what extent *ex vivo* expanded cHSCs could return to quiescence following transplantation, we expanded 100 CD45.2⁺ cHSCs for 21 days and stained all their progeny with the proliferation-tracking dye Cell Trace Violet (CTV) prior to transplantation into CD45.1⁺/CD45.2⁺ hosts. As a control, we transplanted CTV labelled unmanipulated CD45.2⁺ cKit-enriched BM cells (**Fig 5C** [↗](#)). In these animals, splenic CD4⁺ enriched cells were co-transplanted as a control for non-dividing cells (**Fig S5D**). Reconstitution and cHSC fates were monitored biweekly for up to 8-weeks post-transplant. Although the phenotypic cHSC contribution in recipients receiving *ex vivo* expanded cells outnumbered the contribution from unmanipulated cells two-weeks post-transplantation, the levels became comparable later (**Fig 5D** [↗](#)). Recipients from both groups showed high levels of CTV signal two-weeks post-transplantation, with the majority of donor EPCR⁺ cHSCs having undergone fewer than 3 cell divisions. Over time, increasing numbers of cHSCs had proliferated within both groups and with similar patterns (**Fig 5E-F** [↗](#), **Fig S5E**), with a slightly higher number of undivided cHSCs retrieved from *ex vivo* expanded HSCs at all time points (**Fig 5E** [↗](#)). The most quiescent cHSCs associated with higher EPCR levels (**Fig S5F-G**), again attesting to the usefulness of EPCR to predict robust HSC activity.

Therefore, *ex vivo* expanded cHSCs can efficiently engraft immunologically matched hosts without conditioning. Under such conditions, expanded cHSCs can rapidly return to quiescence following transplantation.

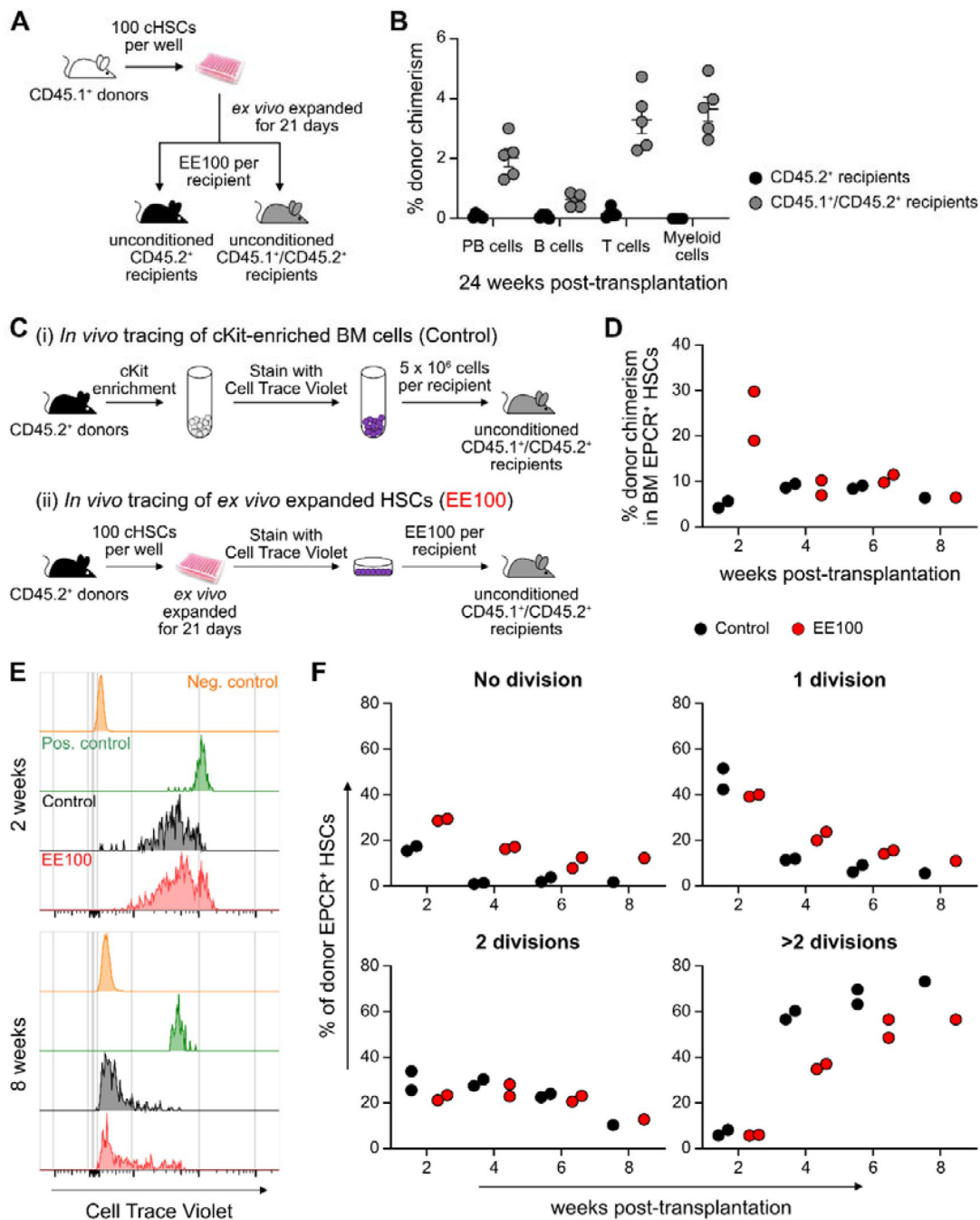


Figure 5.

Cultured cHSCs allow for transplantation into non-conditioned syngeneic recipients.

- (A) Strategy to assess the ability of *ex vivo* expanded cHSCs to engraft unconditioned recipients.
- (B) Test-cell derived PB reconstitution 24-weeks post-transplantation. Symbols denote individual mice and means $\pm$ SEM.
- (C) Strategy used to assess the *in vivo* proliferation of *ex vivo* expanded cHSCs.
- (D) BM cHSC chimerism 2-8 weeks post transplantation.
- (E) Representative CTV label profiles of donor EPCR⁺ cHSCs compared to negative control signal (host EPCR⁺ cHSCs) and positive signal (donor CD4⁺ spleen cells) at 2 or 8 weeks post-transplantation.
- (F) Donor EPCR⁺ cHSCs were evaluated for the number of cell divisions they had undergone through 8 weeks post-transplantation.
- All data points depict values in individual recipients.

Discussion

Although many efforts have aimed to *in vitro* expand murine HSCs, the results have for the most part been disappointing²⁷. A recent study suggested that the substitution of undefined serum products with the synthetic polymer PVA is one key determinant towards this goal³. Here, we detailed this culture system further. We focused on the requirements of candidate input HSCs, the reproducibility of the system, the cellular output and the functional *in vivo* performance of the *in vitro* replicating HSCs. Several observations that we believe are valuable not only for future applications of the PVA culture system, but also for HSC biology in general, emerged from our work.

While the PVA-system supports HSC activity even from unfractionated BM cell preparations²⁸, the long-term repopulating HSCs (LT-HSCs) in cruder input cell preparations is very low. While having its own advantages, this makes it difficult to track the fates of cHSCs. Therefore, we here consistently initiated cultures with more stringently purified cHSCs. As a starting point, we used cHSCs isolated based on a current standard SLAM phenotype (Lin⁻Sca⁺cKit⁺CD48⁻CD150⁺)²⁹. In agreement with previous reports³⁰, we observed that CD201/EPCR further enriches for LT-HSCs, a knowledge that has so far not been adopted as a standard in the field²⁹. This applied both to cHSCs isolated directly from the BM and following culture. In fact, despite that HSCs change many phenotypes in culture⁴, the LT-HSC activity following culture was remarkably enriched not only in the EPCR positive fraction, but also for other phenotypic attributes of unmanipulated LT-HSCs. Importantly, such cHSCs represented only a very minor fraction in cultures (0.6% and 0.1% of 2- and 3-week cultures, respectively) but contained all LT-HSC activity, with the massive overall expansion in cultures leading to substantial net expansions of such cells (100- to 1,200-fold).

While we gained insights into the dynamics of *ex vivo* differentiation of cHSCs into mature lineages by simultaneously measuring ATAC signal and RNA transcripts in individual nuclei using single-cell multiome sequencing, the limited sensitivity of this method hindered the identification of more homogeneous populations of bona fide HSCs within the sorted EPCR⁺ cells. Therefore, we assessed the functional *in vivo* HSC activity in most of our subsequent work. While our clonal barcoding experiments unequivocally demonstrated HSC self-renewal, the large spectra in expansion potential of individual cHSCs was noteworthy. This variation was reduced when assessing HSC activity following culture and aligns with the interpretation that the offspring of cHSCs that self-renew robustly in cultures exhibit less differences. Nonetheless, the demonstrated *in vitro* self-renewal refutes alternative views in which the culture process would mainly act to enhance the reconstitution capacity of individual HSCs, for instance by altering their homing properties and/or by the conversion of more differentiated progenitors into cells with LT-HSC activity. Therefore, our data support the view that this (or other) culture systems might not support all types of LT-HSCs previously alluded to³¹. Further support to this notion can be derived from our data on FL-cHSCs, which did not expand better than adult HSCs in the PVA system, despite that FL HSCs repopulate mice more efficiently than adult HSCs when unmanipulated³². In a broader sense, this associates also with the general concept of self-renewal and that has recently been highlighted in studies on native hematopoiesis, where distinct but only slowly contributing LT-HSCs co-exist with more active progenitors that can presumably also execute self-renewal divisions^{33–35}. It appears that many aspects of this structure are re-created in cultures and, as we show, this requires as input the normally very slowly dividing and presumably most primitive LT-HSCs.

While the parallel self-renewal and differentiation in HSC cultures represent a fundamental difference compared to other *in vitro* stem cell self-renewal systems (e.g., for ES/iPS cells), where preferential self-renewal can be achieved, the differentiation of HSCs can be taken advantage of. We observed that even single cultured HSCs can enhance the speed of donor reconstitution and alleviate the effects of myeloablative conditioning, and the parallel self-renewal and differentiation can also be harnessed to approach HSC fate decisions at a molecular level³⁶. This

was exemplified by our single-cell multimodal experiments, which allowed for determinations of the differentiation trajectories of cHSCs in the culture system and their associated molecular features. However, as the expansion of genuine LT-HSC activity is far from unlimited, a shortage of HSCs is by all likelihood still an obstacle for larger-scale screening efforts such as genome-wide CRISPR screens and larger small molecule screens.

During our studies, we encountered several aspects of the CRA that are often neglected. First, the HSC activity when transplanted at limited numbers highlighted large variations in between recipients, which is in line with previous extensive single cell transplantation experiments^{37,41–42}. While raised previously⁴², taking into account not only the repopulating activity but also clonal aspects has not reached wide use. Emerging barcoding technologies, by which many clones can be evaluated in a single mouse, makes this easier. Secondly, the progeny from a relatively small number of cultured cHSCs (10-50 cells) vastly outcompeted “standard” doses of competitor cells, making quantification unreliable/impossible. This contrasted results from freshly isolated cHSCs transplanted at the same doses used to initiate cultures. While this concern could be overcome by adjusting the graft composition (e.g. enhancing the number of competitor cells), extensive variation still existed among cultures that could be eliminated by using larger amounts of input cells and splitting the contents into individual recipients. This should have relevance also for normal HSC biology by highlighting intrinsic rather than stochastic regulation as the central determinant for persistent HSC function. A third consideration relates to the definition of ongoing HSC activity. We and others have previously established that this is best mirrored by the HSC contribution to myelopoiesis^{43,44}. When we assessed cultured EPCR⁺CD48⁺ cells (with predominantly transient multilineage activity), we observed prominent long-term T cell contribution but limited myeloid reconstitution. This should not be confused with the concept of HSC lineage-bias, but rather reflects that the pool of T cells *in vivo* can be maintained by homeostatic proliferation and thus do not require continuous input from HSCs. Thus, short-lived non-self-renewing multipotent progenitors, which are generated in large numbers in culture, can contribute extensive (lymphoid) offspring in the long-term but should not be mistaken for ongoing HSC activity. This consideration is valid also for competitor grafts in the CRA (in the form of WBM cells), which contain numerous progenitor cells that can rapidly and persistently contribute to lymphopoiesis. Thus, as opposed to the classic CRA, where the overall donor contribution is assessed⁹, the contribution to short-lived lineages reflects better the ongoing HSC activity, with the contribution to lymphoid lineages serving as a qualitative parameter for multipotency. In agreement with previous viral barcoding studies on HSCs¹⁴, we observed that when cultured HSCs are barcoded and assessed long-term, there is a strong correlation between barcodes in BM progenitors and myeloid cells, while the pool of mature B cells contain clones that cannot be recovered in other lineages (data not shown).

Successful repopulation of non-conditioned hosts is based on the suggestion that large numbers of HSCs are needed to saturate those few BM niches available for engraftment^{18,45}. Avoiding toxic myeloablation has apart from evident clinical implications also a precedence in experimental murine HSC biology, where lethal myeloablation enforces vigorous HSC proliferation following transplantation¹², but which contrasts native contexts^{12,33,34}. We observed that the CD45.1/CD45.2 differences are sufficiently immunogenic to mediate long-term graft failure in the non-conditioned setting, which in hindsight has been alluded to previously^{18,26,46}. Thus, by matching hosts for CD45-isoforms, we could successfully obtain low-level (2-5%) long-term multilineage chimerism in adult non-myeloablated hosts from only 100 cHSCs. Intriguingly, despite being activated to proliferate *in vitro*, many cHSCs rapidly returned to a quiescent state *in vivo*, with the most dormant cHSCs exhibiting the most stringent HSC phenotype (EPCR^{high} SLAM LSKs). The possibility for non-conditioned transplantation should complement genetic lineage tracing models aimed at exploring both normal and malignant hematopoiesis in more native/physiological contexts^{12,33,34}.

In summary, we here characterized and detailed murine HSCs as they self-renew and differentiate *in vitro*. Although several aspects of the PVA system remain to be explored, a powerful *in vitro* self-renewal system has been a long-sought goal in HSC biology where HSCs have been notoriously difficult to uphold in an undifferentiated/self-renewal state. The merger of this system with recent advances in transplantation biology, genome engineering and single-cell techniques holds promise for many exciting discoveries of relevance to both basic and more clinically oriented hematopoietic research.

Materials and methods

Mice

Adult (8-10-week-old) C57Bl/6N (CD45.2⁺) female mice were purchased from Taconic. Transgenic Fgd5-ZsGreen-2A-CreERT2 mice²² (JAX:027789) was a kind gift from Derrick Rossi. Mice were maintained in the animal facilities at the Biomedical Center of Lund University and kept in environment-enriched conditions with 12-hour light-dark cycles and water and food provided *ad libitum*. All experimental procedures were approved by a local ethical committee (permits M186-15 and 16468/2020). All efforts were made to reduce the number of experimental animals and suffering.

BM transplantation

All mice used as recipients were 8 - 12 weeks of age. For conditioned recipients, mice were lethally irradiated (950 cGy) at least 4 h prior transplantation. The conditioned mice received prophylactic ciprofloxacin (HEXAL, 125 mg/L in drinking water) for 2 weeks beginning on the day of irradiation. All transplantations were performed through intravenous tail vein injection. The amount of transplanted test cells and competitor/support cells is described in the results section.

In vitro HSC culture

In vitro HSC cultures were performed using F12-PVA based culture conditions as previously described³. In brief, cHSCs (HSCs (EPCR^{high}CD150⁺CD48^{low} LSKs) were sorted into 96-well flat-bottom plates (Thermo Scientific) coated with 100 ng/ml fibronectin (Sigma) for >1 hour at 37°C prior to sorting. HSCs were sorted into 200 µl HSC media (Supplemental Table 1) and expanded at 37°C with 5% CO₂ for up to 21 days. The first media changes were performed 5 days (for ≥ 50 initial HSCs), 10 days (for 10 initial HSCs) or 15 days (for single-cell cultures) after initiation of cultures and then every 2 days. For pre-cultures associated with lentiviral transduction, the first media change was performed 24h after transduction and then as above. Cells were split at a 1:2-1:4 ratio into new fibronectin coated plates when reaching 80-90% confluency (normally every 4 days after the first split). In order to expand cHSCs to reach a workable number as well as avoiding vast numbers of differentiated cells generated after very long-term of culture (f.i. up to 4 weeks), the cHSCs were kept expanding *ex vivo* for 3 weeks as a standard protocol. Following 14- and/or 21-day expansion, cellularity was assessed using an Automated Cell Counter (TC20, BioRad) and used for flow cytometric analyses and/or transplantation. BM cells collected from wild type animals were pooled together for sorting to initiate HSC cultures. Each initial well was considered as a technical replicate and treated/analyzed separately.

Cell preparation

Mice were euthanized by cervical dislocation after which bones (both right and left tibias, femurs, and pelvis) or spleens were extracted. Fetal cells were extracted from livers at E14.5. Bones were crushed using a mortar and pestle and BM cells were collected in ice-cold phosphate-buffered saline (PBS, Gibco) with 1% fetal bovine serum (Sigma-Aldrich) (FACS buffer), filtered (100 µm mesh) and washed. FL or spleen cells were brought into single cell suspension by grinding through

a 100 μ m mesh. BM or FL cells were cKit-enriched by anti-cKit-APCeFluor780 (eBioscience) staining and spleen cells were CD4-enriched by anti-CD4-APC-Cy7 staining, followed by incubation with anti-APC conjugated beads (Miltenyi Biotec). Magnetic separation was performed using LS columns according to manufacturer's instructions (Miltenyi Biotec). cKit-enriched BM or FL cells were washed and stained with fluorescently labeled antibodies for analysis or sorting.

Peripheral blood (PB) chimerism analysis after transplantation was done as previously described³⁵. In brief, blood was drawn from tail vein into FACS buffer containing 10 U/ml Heparin (Leo). After incubating with an equal volume of 2% dextran (Sigma) in PBS at 37 °C, the upper phase was collected and erythrocytes were lysed using an ammonium chloride solution (STEMCELL Technologies) for 3 min at room temperature, followed by washing.

For analysis and/or sorting after *ex vivo* expansion, cultured cells were resuspended by pipetting and collected using FACS buffer. An aliquot was taken for cell counting using an Automated Cell Counter (TC20, BioRad).

Flow cytometry analysis and FACS sorting

Cells were kept on ice when possible, with the FACS buffer kept ice-cold. Staining, analysis and sorting were performed as previously described¹². In brief, cells were resuspended in Fc-block (1:50, 5 x 10⁶ cells/50 μ l, InVivoMab) for 10 min and then for 20 min with a twice concentrated antibody staining mixture (Supplemental Tables 2-10) at 4LJ°C in dark. In case biotinylated lineage markers were included, a secondary staining with streptavidin BV605 was performed for 20 min (1:400, 5 x 10⁶ cells/100 μ l, Sony) at 4LJ°C in dark. After a final wash, the cells were resuspended in FACS buffer containing propidium iodide (1:1000, Invitrogen).

All FACS experiments were performed at the Lund Stem Cell Center FACS Core Facility (Lund University) on Aria III and Fortessa X20 instruments (BD). Bulk populations were sorted using a 70-micron nozzle, 0.32.0 precision mask, and a flow rate of 2 - 3K events/s. HSCs for single cell culture were index sorted. FACS analysis was performed using FlowJo version 10.8.0 (Tree Star Inc.).

Multome single-cell sequencing

Library preparation

Multome sequencing experiments were performed at the Center for Translational Genomics (Lund University) using the Chromium Next GEM Single Cell Multome ATAC + Gene Expression Reagent Bundle kit according to manufacturer's instructions (10x Genomics). 40,000 viable cells or EPCR⁺ viable cells were sorted from *ex vivo* expanded cultures for multome single-cell sequencing. Data has been uploaded to GEO under accession number GSE234906 (<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE234906>).

Bioinformatic analysis

Data was mainly analyzed with the Seurat⁴⁷ and Signac⁴⁸ R⁴⁹ packages, as well as the SAILERX Python package⁵⁰ for the joint-modality dimensionality reduction. All accompanying code for the single-cell multome analysis post Cell Ranger can be found at this Github repository (https://github.com/razofz/DB_QZ_multome), including conda specifications for the environments used with exact versions of all packages.

i. Preprocessing

The count matrices were generated with the Cell Ranger ARC tool (v2.0.0), aligned to the reference genome mm10. The aggregated dataset (containing data for both samples) were then loaded into Signac/Seurat in R where non-standard chromosomes were removed and the Seurat object split into the two samples, here named Diverse and Immature, which corresponds to whole culture or sorted EPCR⁺ cells, respectively. The data were then filtered on UMI, gene and peak counts, as well as nucleosome signal and TSS enrichment, with separate thresholds for the two samples. The ATAC features were classified as either distal or proximal peaks, and split accordingly. Proximal peaks were defined as a ± 2 kbp region around the transcription start site (TSS). The TSS positions were retrieved from GREAT's website, and GREAT v4 mm10 version was used. All peaks not classified as proximal received the classification "distal". Motifs were then identified for distal and proximal peaks separately⁵¹ [\[link\]](#).

ii. Downstream processing and joint-modality dimensionality reduction

The samples were processed separately. Since SAILERX uses pre-existing dimensionality reduction and clusters for the RNA modality, the RNA data was normalized and scaled, highly variable genes (HVGs) were found, principal component analysis (PCA) was run and clustering was performed, all with the Seurat standard functions with default parameters. After that, relevant data was extracted (metadata including RNA clusters, PCA embeddings, a whitelist of non-filtered out cell barcodes, ATAC peak count matrix) and with it an hdf5 file similar to the example files provided by the SAILERX repository was created (<https://github.com/uci-cbcl/SAILERX> [\[link\]](#)). A SAILERX model was then trained on the generated hdf5 files (still sample-wise) according to their provided instructions. The resulting embeddings from the SAILERX joint-modality dimensionality were then exported and imported into the respective Seurat objects. In Seurat, the embeddings were used for graph construction, clustering and constructing UMAPs. Scores for cell cycle gene signatures were calculated and visualized (**Fig 2E** [\[link\]](#) for the Immature sample) using Seurat's corresponding standard functions with default parameters.

iii. Differential gene expression

Differential gene expression testing was performed on the resulting clusters using Seurat's FindAllMarkers function with the HVGs found above. This analysis was used to calculate HSC signature score as described below (v. HSC signature).

iv. ATAC processing and cluster annotation

Motif activity scores were identified with chromVAR⁵² [\[link\]](#) for distal and proximal peaks respectively, still sample-wise. Differentially active motifs (DAMs) were then identified for each cluster. Annotation of clusters in terms of cell type was then performed manually with the help of DEGs and DAMs (**Fig 2B** [\[link\]](#)), guided by preexisting data on the expression profiles of cell types using the Enrichr platform (<https://maayanlab.cloud/Enrichr/> [\[link\]](#)).

v. HSC signature

A 13 gene condensed HSC signature list (Fig S2C) was obtained by mining genes associated to cluster 5 with publicly available gene expression data of HSCs from the ImmGen consortia and Bloodspot (<https://servers.binf.ku.dk/bloodspot/> [\[link\]](#)). This signature was then used to calculate a score with Seurat's AddModuleScore function. A classification for expression of this gene signature was performed with a slightly modified version of Seurat's CellCycleScoring function, where instead of three classifications (S, G2M or Undecided/G1) two classifications were used (expressed or not expressed). This classification was then used to create the contours used in **Fig 2D-E** [\[link\]](#), and the signature score used for coloring the points in the same plots.

vi. Trajectory inference in the Diverse sample

Trajectory inference was performed for the Diverse sample with the slingshot package⁵³. As initiating cluster, the cluster 5 was assigned (the earliest cluster), and end clusters chosen were clusters 4, 1, 6 and 7. The slingshot function of slingshot was used with otherwise default parameters. A joint UMAP with the four resulting trajectories was constructed (Fig 2C). To circumvent the problem of interfering color scales, only the top 20% expressing cells for each marker was chosen for coloring, and overlapping cells between these sets were excluded.

Assessment of HSC heterogeneity via DNA barcoding

Lentiviral barcode libraries were purchased from AddGene (No. 115645 and No. 83993) and amplified according to instructions^{24,25}. AddGene 83993 is a Crispri gRNA library that was repurposed here for the purpose of lentiviral barcoding (e.g. no Cas9 was co-expressed). These libraries were kind gifts from Samantha Morris and Jonathan Weissman.

For pre-culture labelling, five wells containing 1000 CD45.1⁺ HSCs each were sorted and cultured for 48 h. Cells were then transduced with the AddGene 115645 library (containing >5737 unique barcodes) overnight with approximately 35% transduction efficiency, which was followed by regular culture procedures. After 19 days, expanded cells were collected from each initial wells separately. Half of the expanded cells were transplanted into individual ‘parental’ CD45.2⁺ recipients and the rest were mixed together and transplanted into five ‘daughter’ CD45.2⁺ recipients. Each ‘parental’ or ‘daughter’ recipient received an estimated expansion equivalent to 500 cHSCs. For post-culture labelling, 3000 cHSCs were sorted from CD45.1⁺ mice and expanded for 13 days. Thereafter, the cells were transduced with the AddGene 83993 library (containing >10090 unique barcodes) overnight at an approximate density of 100,000 cells/well. Half of the cultured and transduced cells were transplanted into 5 CD45.2⁺ WT recipients. Each recipient received an estimated expansion equivalent of 300 cHSCs, of which approximately 15% were transduced according to assessments of transduction efficiencies. Barcode labelled cells were collected and analyzed from four recipients. For both pre- and post-culture labeling experiments, recipients were lethally irradiated and provided with 500,000 unfractionated CD45.2⁺ WBM cells each as support. At the experimental end points (16 weeks after transplantation), barcode labelled BM myeloid cells were sorted into 200 µl RNA lysis buffer (Norgen Biotek Corp) (“pre-culture” barcoding) or pelleted for genomic DNA extraction (“post-culture” barcoding) and proceeded to library preparation and sequencing analysis.

RNA was isolated according to the protocol of Single Cell RNA Purification Kit (Norgen Biotek Corp) and cDNA were synthesis using qScript cDNA SuperMix (Quantabio). Genomic DNA was extracted using PureLink Genomic DNA Mini Kit (Invitrogen). The 8-(pre-culture) and 21-base pair (bp) barcodes (post-culture) were amplified by PCR using Q5 High-Fidelity DNA Polymerase (New England Biolabs). Primers used are listed in Supplemental Table 11. Adapters containing index sequences (Supplemental Table 10) were added by PCR for 7 cycles. After each PCR amplification step, the products were cleaned up using SPRIselect beads (Beckman Coulter). Purified libraries were run on Agilent High Sensitivity DNA Kit chip (Agilent Technologies) to verify the expected size distribution, quantified by Qubit dsDNA HS Assay Kit (Thermo Fisher Scientific), and pooled at equimolar concentrations. Pooled libraries were loaded on an Illumina MiSeq Reagent Nano Kit v2 flow cell following protocols for single-end 100-cycle sequencing. FASTQ files were further assessed with FastQC (v0.11.2), including read count, base quality across reads, and guanine and cytosine (GC) content per sequence.

After extracting the reads of each barcode, the frequency in each recipient was calculated and barcodes with a frequency less than 0.1% were considered background reads. This is an arbitrary cut-off. Barcodes present in more than one ‘parental’ recipient were excluded from analysis (7 %

and 2% of barcodes pre- and post-culture, respectively) to avoid the possibility of including HSCs transduced independently with the same barcode or potential over-represented barcodes. Reads were normalized to 10^6 total reads per recipient followed by statistical analyses.

Cell Trace Violet labelling for *in vitro* and *in vivo* HSC proliferation tracing

CD45.2⁺ cultured cells, cKit-enriched BM cells or CD4-enriched spleen cells were collected, washed and pelleted. Cells were resuspended to a final concentration of 5×10^6 cells/ml and labelled with 1 μ M CTV (Invitrogen) in PBS for 10 min at 37°C. The reaction was stopped by adding same volume of ice-cold FACS buffer and washing again with FACS buffer. For *in vitro* proliferation tracing, 10^5 CTV labeled cKit-enriched BM cells were plated into each well containing F12-PVA culture media (Supplemental Table 1). At each time point, half of the cultured cells were collected and stained for evaluating cHSC CTV signals (Supplemental Table 8) while the remaining cells were kept in culture for continue expansion. For *in vivo* proliferation tracing, CTV labeled EE100 cultured cells or 5×10^6 cKit-enriched BM cells were mixed with 2×10^6 CTV labeled CD4-enriched spleen cells and transplanted into one CD45.1⁺/CD45.2⁺ unconditioned recipients. For analysis, BM cells were cKit-enriched and stained accordingly (Supplemental Table 9). CD4 positive spleen cells were collected from the same recipient as a control for positive signal/undivided cells (Supplemental Table 10).

Statistical test

Statistical analyses were performed using Microsoft Excel and Graphpad Prism 9. Significance was calculated by Mann-Whitney tests if not otherwise specified. Statistical significances are described in figure legends for the relevant graphs. In all legends, n denotes biological replicates.

Acknowledgements

We acknowledge Dr. Shabnam Kharazi for scientific discussions and technical support, Mr. Yun Sheng for advice on analysis for barcode sequencing and Prof. Hiromitsu Nakauchi and Dr. Adam C. Wilkinson for advice on cell cultures. The work was supported by grants from the Tobias Foundation, the Swedish Cancer Foundation, Barncancerfonden and the Swedish Research Council to D.B. and the Royal Physiographic Society of Lund foundation to Q.Z and A.K-C.

Authorship contributions

Q.Z. designed the research, performed experiments, analyzed data, and wrote the manuscript; R.O. performed bioinformatic analysis and wrote the manuscript; A.K-C. and O.Y. performed experiments and analyzed data; D.B. supervised the project, was responsible for funding acquisition, designed the research, analyzed data and wrote the manuscript.

Conflict of interest disclosures

The authors report no conflict of interest.

References

- 1 Konturek-Ciesla A., Bryder D. (2022) **Stem Cells, Hematopoiesis and Lineage Tracing: Transplantation-Centric Views and Beyond** *Frontiers in Cell and Developmental Biology* **10** <https://doi.org/10.3389/fcell.2022.903528>
- 2 Walasek M. A., van Os R., de Haan G (2012) **Hematopoietic stem cell expansion: challenges and opportunities** *Ann N Y Acad Sci* **1266**:138–150 <https://doi.org/10.1111/j.1749-6632.2012.06549.x>
- 3 Wilkinson A. C., et al. (2019) **Long-term ex vivo haematopoietic-stem-cell expansion allows nonconditioned transplantation** *Nature* <https://doi.org/10.1038/s41586-019-1244-x>
- 4 Zhang C. C., Lodish H. F (2005) **Murine hematopoietic stem cells change their surface phenotype during ex vivo expansion** *Blood* **105**:4314–4320 <https://doi.org/10.1182/blood-2004-11-4418>
- 5 Nakauchi H., Takano H., Ema H., Osawa M (1999) **Further characterization of CD34-low/negative mouse hematopoietic stem cells** *Ann N Y Acad Sci* **872**:57–66 <https://doi.org/10.1111/j.1749-6632.1999.tb08453.x>
- 6 Yang L., et al. (2005) **Identification of Lin(-)Sca1(+)kit(+)CD34(+)Flt3-short-term hematopoietic stem cells capable of rapidly reconstituting and rescuing myeloablated transplant recipients** *Blood* **105**:2717–2723 <https://doi.org/10.1182/blood-2004-06-2159>
- 7 Nakorn Na, Traver T., Weissman D., Akashi I. L. (2002) **Myeloerythroid-restricted progenitors are sufficient to confer radioprotection and provide the majority of day 8 CFU-S** *J Clin Invest* **109**:1579–1585 <https://doi.org/10.1172/JCI15272>
- 8 Purton L. E., Scadden D. T (2007) **Limiting factors in murine hematopoietic stem cell assays** *Cell Stem Cell* **1**:263–270 <https://doi.org/10.1016/j.stem.2007.08.016>
- 9 Harrison D. E., Jordan C. T., Zhong R. K., Astle C. M (1993) **Primitive hemopoietic stem cells: direct assay of most productive populations by competitive repopulation with simple binomial, correlation and covariance calculations** *Exp Hematol* **21**:206–219
- 10 Barnett D., et al. (1999) **Guideline for the flow cytometric enumeration of CD34+ haematopoietic stem cells. Prepared by the CD34+ haematopoietic stem cell working party. General Haematology Task Force of the British Committee for Standards in Haematology** *Clin Lab Haematol* **21**:301–308 <https://doi.org/10.1046/j.1365-2257.1999.00253.x>
- 11 Trebeden-Negre H., et al. (2010) **Delayed recovery after autologous peripheral hematopoietic cell transplantation: potential effect of a high number of total nucleated cells in the graft** *Transfusion* **50**:2649–2659 <https://doi.org/10.1111/j.1537-2995.2010.02746.x>
- 12 Sawen P., et al. (2016) **Mitotic History Reveals Distinct Stem Cell Populations and Their Contributions to Hematopoiesis** *Cell Rep* **14**:2809–2818 <https://doi.org/10.1016/j.celrep.2016.02.073>

- 13 Bryder D., Sasaki Y., Borge O. J., Jacobsen S. E (2004) **Deceptive multilineage reconstitution analysis of mice transplanted with hemopoietic stem cells, and implications for assessment of stem cell numbers and lineage potentials** *J Immunol* **172**:1548–1552 <https://doi.org/10.4049/jimmunol.172.3.1548>
- 14 Lu R., Neff N. F., Quake S. R., Weissman I. L (2011) **Tracking single hematopoietic stem cells in vivo using high-throughput sequencing in conjunction with viral genetic barcoding** *Nat Biotechnol* **29**:928–933 <https://doi.org/10.1038/nbt.1977>
- 15 Busch K., Rodewald H. R (2016) **Unperturbed vs. post-transplantation hematopoiesis: both in vivo but different** *Curr Opin Hematol* **23**:295–303 <https://doi.org/10.1097/MOH.0000000000000250>
- 16 Brecher G., Ansell J. D., Micklem H. S., Tjio J. H., Cronkite E. P (1982) **Special proliferative sites are not needed for seeding and proliferation of transfused bone marrow cells in normal syngeneic mice** *Proc Natl Acad Sci U S A* **79**:5085–5087 <https://doi.org/10.1073/pnas.79.16.5085>
- 17 Bhattacharya D., et al. (2009) **Niche recycling through division-independent egress of hematopoietic stem cells** *J Exp Med* **206**:2837–2850 <https://doi.org/10.1084/jem.20090778>
- 18 Bhattacharya D., Rossi D. J., Bryder D., Weissman I. L (2006) **Purified hematopoietic stem cell engraftment of rare niches corrects severe lymphoid deficiencies without host conditioning** *J Exp Med* **203**:73–85 <https://doi.org/10.1084/jem.20051714>
- 19 Balazs A. B., Fabian A. J., Esmon C. T., Mulligan R. C (2006) **Endothelial protein C receptor (CD201) explicitly identifies hematopoietic stem cells in murine bone marrow** *Blood* **107**:2317–2321 <https://doi.org/10.1182/blood-2005-06-2249>
- 20 Gekas C., Graf T (2013) **CD41 expression marks myeloid-biased adult hematopoietic stem cells and increases with age** *Blood* **121**:4463–4472 <https://doi.org/10.1182/blood-2012-09-457929>
- 21 Wilkinson A. C., Ishida R., Nakauchi H., Yamazaki S (2020) **Long-term ex vivo expansion of mouse hematopoietic stem cells** *Nat Protoc* <https://doi.org/10.1038/s41596-019-0263-2>
- 22 Gazit R., et al. (2014) **Fgd5 identifies hematopoietic stem cells in the murine bone marrow** *J Exp Med* **211**:1315–1331 <https://doi.org/10.1084/jem.20130428>
- 23 Domen J., Cheshier S. H., Weissman I. L (2000) **The role of apoptosis in the regulation of hematopoietic stem cells: Overexpression of Bcl-2 increases both their number and repopulation potential** *J Exp Med* **191**:253–264 <https://doi.org/10.1084/jem.191.2.253>
- 24 Bidy B. A., et al. (2018) **Single-cell mapping of lineage and identity in direct reprogramming** *Nature* **564**:219–224 <https://doi.org/10.1038/s41586-018-0744-4>
- 25 Horlbeck M. A., et al. (2016) **Compact and highly active next-generation libraries for CRISPR-mediated gene repression and activation** *Elife* **5** <https://doi.org/10.7554/eLife.19760>
- 26 van Os R., et al. (2001) **Immunogenicity of Ly5 (CD45)-antigens hampers long-term engraftment following minimal conditioning in a murine bone marrow transplantation model** *Stem Cells* **19**:80–87 <https://doi.org/10.1634/stemcells.19-1-80>

- 27 Wilkinson A. C., Igarashi K. J., Nakauchi H (2020) **Haematopoietic stem cell self-renewal in vivo and ex vivo** *Nature reviews. Genetics* **21**:541–554 <https://doi.org/10.1038/s41576-020-0241-0>
- 28 Ochi K., Morita M., Wilkinson A. C., Iwama A., Yamazaki S (2021) **Non-conditioned bone marrow chimeric mouse generation using culture-based enrichment of hematopoietic stem and progenitor cells** *Nat Commun* **12** <https://doi.org/10.1038/s41467-021-23763-z>
- 29 Challen G. A., Pietras E. M., Wallscheid N. C., Signer R. A. J (2021) **Simplified murine multipotent progenitor isolation scheme: Establishing a consensus approach for multipotent progenitor identification** *Exp Hematol* **104**:55–63 <https://doi.org/10.1016/j.exphem.2021.09.007>
- 30 Kent D. G., et al. (2009) **Prospective isolation and molecular characterization of hematopoietic stem cells with durable self-renewal potential** *Blood* **113**:6342–6350 <https://doi.org/10.1182/blood-2008-12-192054>
- 31 Haas S., Trumpp A., Milsom M. D (2018) **Causes and Consequences of Hematopoietic Stem Cell Heterogeneity** *Cell Stem Cell* **22**:627–638 <https://doi.org/10.1016/j.stem.2018.04.003>
- 32 Rebel V. I., Miller C. L., Eaves C. J., Lansdorp P. M (1996) **The repopulation potential of fetal liver hematopoietic stem cells in mice exceeds that of their liver adult bone marrow counterparts** *Blood* **87**:3500–3507
- 33 Sun J., et al. (2014) **Clonal dynamics of native haematopoiesis** *Nature* **514**:322–327 <https://doi.org/10.1038/nature13824>
- 34 Busch K., et al. (2015) **Fundamental properties of unperturbed haematopoiesis from stem cells in vivo** *Nature* **518**:542–546 <https://doi.org/10.1038/nature14242>
- 35 Sawen P., et al. (2018) **Murine HSCs contribute actively to native hematopoiesis but with reduced differentiation capacity upon aging** *Elife* **7** <https://doi.org/10.7554/eLife.41258>
- 36 Weinreb C., Rodriguez-Fraticelli A., Camargo F. D., Klein A. M (2020) **Lineage tracing on transcriptional landscapes links state to fate during differentiation** *Science* **367** <https://doi.org/10.1126/science.aaw3381>
- 37 Yamamoto R., et al. (2018) **Large-Scale Clonal Analysis Resolves Aging of the Mouse Hematopoietic Stem Cell Compartment** *Cell Stem Cell* **22**:600–607 <https://doi.org/10.1016/j.stem.2018.03.013>
- 38 Yamamoto R., et al. (2013) **Clonal analysis unveils self-renewing lineage-restricted progenitors generated directly from hematopoietic stem cells** *Cell* **154**:1112–1126 <https://doi.org/10.1016/j.cell.2013.08.007>
- 39 Benz C., et al. (2012) **Hematopoietic stem cell subtypes expand differentially during development and display distinct lymphopoietic programs** *Cell Stem Cell* **10**:273–283 <https://doi.org/10.1016/j.stem.2012.02.007>
- 40 Carrelha J., et al. (2018) **Hierarchically related lineage-restricted fates of multipotent haematopoietic stem cells** *Nature* **554**:106–111 <https://doi.org/10.1038/nature25455>

- 41 Oguro H., Ding L., Morrison S. J (2013) **SLAM family markers resolve functionally distinct subpopulations of hematopoietic stem cells and multipotent progenitors** *Cell Stem Cell* **13**:102–116 <https://doi.org/10.1016/j.stem.2013.05.014>
- 42 Ema H., Nakauchi H (2000) **Expansion of hematopoietic stem cells in the developing liver of a mouse embryo** *Blood* **95**:2284–2288
- 43 Norddahl G. L., et al. (2011) **Accumulating mitochondrial DNA mutations drive premature hematopoietic aging phenotypes distinct from physiological stem cell aging** *Cell Stem Cell* **8**:499–510 <https://doi.org/10.1016/j.stem.2011.03.009>
- 44 Domen J., Weissman I. L (2003) **Hematopoietic stem cells and other hematopoietic cells show broad resistance to chemotherapeutic agents in vivo when overexpressing bcl-2** *Exp Hematol* **31**:631–639 [https://doi.org/10.1016/s0301-472x\(03\)00084-5](https://doi.org/10.1016/s0301-472x(03)00084-5)
- 45 Czechowicz A., Kraft D., Weissman I. L., Bhattacharya D (2007) **Efficient transplantation via antibody-based clearance of hematopoietic stem cell niches** *Science* **318**:1296–1299 <https://doi.org/10.1126/science.1149726>
- 46 Xu H., Exner B. G., Chilton P. M., Schanie C., Ildstad S. T (2004) **CD45 congenic bone marrow transplantation: evidence for T cell-mediated immunity** *Stem Cells* **22**:1039–1048 <https://doi.org/10.1634/stemcells.22-6-1039>
- 47 Hao Y., et al. (2021) **Integrated analysis of multimodal single-cell data** *Cell* **184**:3573–3587 <https://doi.org/10.1016/j.cell.2021.04.048>
- 48 Stuart T., Srivastava A., Madad S., Lareau C. A., Satija R (2021) **Single-cell chromatin state analysis with Signac** *Nat Methods* **18**:1333–1341 <https://doi.org/10.1038/s41592-021-01282-5>
- 49 Team R. C. (2022) **Team, R. C. R: A language and environment for statistical computing. (2022).** <https://www.R-project.org/>
- 50 Cao Y., et al. (2022) **Integrated analysis of multimodal single-cell data with structural similarity** *Nucleic acids research* **50** <https://doi.org/10.1093/nar/gkac781>
- 51 Fornes O., et al. (2020) **JASPAR 2020: update of the open-access database of transcription factor binding profiles** *Nucleic acids research* **48**:D87–D92 <https://doi.org/10.1093/nar/gkz1001>
- 52 Schep A. N., Wu B., Buenrostro J. D., Greenleaf W. J (2017) **chromVAR: inferring transcription-factor-associated accessibility from single-cell epigenomic data** *Nat Methods* **14**:975–978 <https://doi.org/10.1038/nmeth.4401>
- 53 Street K., et al. (2018) **Slingshot: cell lineage and pseudotime inference for single-cell transcriptomics** *BMC Genomics* **19** <https://doi.org/10.1186/s12864-018-4772-0>

Article and author information

Qinyu Zhang

Division of Molecular Hematology, Department of Laboratory Medicine, Lund Stem Cell Center, Faculty of Medical, Lund University, 221 84 Lund, Sweden
 ORCID iD: [0000-0001-8292-938X](https://orcid.org/0000-0001-8292-938X)

Rasmus Olofzon

Division of Molecular Hematology, Department of Laboratory Medicine, Lund Stem Cell Center,
Faculty of Medical, Lund University, 221 84 Lund, Sweden
ORCID iD: [0000-0001-8079-8718](https://orcid.org/0000-0001-8079-8718)

Anna Konturek-Ciesla

Division of Molecular Hematology, Department of Laboratory Medicine, Lund Stem Cell Center,
Faculty of Medical, Lund University, 221 84 Lund, Sweden

Ouyang Yuan

Division of Molecular Hematology, Department of Laboratory Medicine, Lund Stem Cell Center,
Faculty of Medical, Lund University, 221 84 Lund, Sweden

David Bryder

Division of Molecular Hematology, Department of Laboratory Medicine, Lund Stem Cell Center,
Faculty of Medical, Lund University, 221 84 Lund, Sweden
For correspondence: David.Bryder@med.lu.se
ORCID iD: [0000-0002-8761-4237](https://orcid.org/0000-0002-8761-4237)

Copyright

© 2023, Zhang et al.

This article is distributed under the terms of the [Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/), which permits unrestricted use and redistribution provided that the original author and source are credited.

Editors

Reviewing Editor

Dominique Bonnet

The Francis Crick Institute, London, United Kingdom

Senior Editor

Didier Stainier

Max Planck Institute for Heart and Lung Research, Bad Nauheim, Germany

Reviewer #1 (Public Review):

In 2019, Wilkinson and colleagues (PMID: 31142833) managed to break the veil on a 20-year open question on how to properly culture and expand Hematopoietic Stem Cells (HSCs). Although this study is revolutionizing the HSC biology field, several questions regarding the mechanisms of expansion remain open. Leveraging on this gap, Zhang et al.; embarked on a much-needed investigation regarding HSC self-renewal in this particular culturing setting.

The authors firstly tackled the known caveat that some HSC membrane markers are altered during in vitro cultures by functionally establishing EPCR (CD201) as a reliable and stable HSC marker (Figure 1), demonstrating that this compartment is also responsible for long-term hematopoietic reconstitution (Figure 3). Next in Figure 2, the authors performed single-cell omics to shed light into the potential mechanisms involved into HSC maintenance, and interestingly it was shown that several hematopoietic populations like monocytes and neutrophils are also present in this culture conditions, which has not been reported. The study goes on to functionally characterize these cultured HSCs (cHSC). The authors elegantly

demonstrate using state-of-the-art barcoding strategies that these culturing conditions provoke heterogeneity in the expanding HSC pool (Figure 4). In the last experiment (Figure 5), it was demonstrated that cHSC not only retain their high EPCR expression levels but upon transplantation these cells remain more quiescent than freshly-isolated controls.

Taken together, this study independently validates that the proposed culturing system works and provide new insights into the mechanisms whereby HSC expansion takes place.

Following a first round of comments, the authors provided a comprehensive point-by-point response to the different points raised by reviewers, which significantly helps on better understanding some of the decisions taken by the authors. However, it is surprising that the current manuscript is practically unchanged compared to the previous version. Effectively, all major comments raised by reviewers are address in the response letter rather than incorporated into a truly updated version, which would be of great benefit for readers.

Further comments:

1. It is highly appreciated that the authors provide a comprehensive and cohesive explanations on i) the rationale for employing SAILERX for single-cell RNA and ATAC-seq, ii) data on HSC signature projected on independent scRNA-seq datasets and iii) further context on the *Egd5* expression limitations. These are important snippets of information which do not only further validate this manuscript's data but also provide context within the HSC biology field.

However, I do not fully agree with the author statement "our primary objective in this study was to highlight the relatively low content of HSCs in cultures" (page 1, response to Reviewers) justifying why single-cell genome-wise approaches were used. As the authors are aware HSCs are defined by functional characterization rather than transcriptional/chromatin accessibility profiles, so it seems odd that this was the rationale to perform omics for this purpose. More importantly, the authors had gone through the lengths of already performing this costly and time-consuming experiment, but miss out on the opportunity to take a deeper dive into the molecular characteristics that could explain divergent behavior between freshly-isolated and cultured HSCs. It would be extremely relevant to the HSC biology community to understand, for example, if these two HSC populations have differences in enhancer accessibility (if the data quality allows), which could provide an upstream explanation for differences in transcription (is also not explored in this manuscript version).

2. It intriguing that the authors acknowledge that there are already more recent versions of this expansion protocol (page 2, response to Reviewers) and provided a convoluted explanation on why these were not included in the original manuscript. Both papers (PMID: 36809781 and PMID: 37385251) have now been published in respected peer-reviewed journals and provide insights which are pertinent for this work. Yet, the authors decided not to discuss these findings. It is understandable that repeating experiments with these updated conditions is outside of the scope of this manuscript, but it would be relevant to discuss how these recent advances in the protocol impact the work presented in this manuscript.

3. Regarding the previous comment on how cultured HSC are related to HSC aging, I highly appreciate both data on serial transplantation and also on scRNA-seq.

<https://doi.org/10.7554/eLife.91826.2.sa1>

Reviewer #2 (Public Review):

Summary:

In this study, Zhang and colleagues characterise the behaviour of mouse hematopoietic stem cells when cultured in PVA conditions, a recently published method for HSC expansion (Wilkinson et al., Nature, 2019), using multiome analysis (scRNA-seq and scATACseq in the

same single cell) and extensive transplantation experiments. The latter are performed in several settings including barcoding and avoiding recipient conditioning. Collectively the authors identify several interesting properties of these cultures namely: 1) only very few cells within these cultures have long-term repopulation capacity, many others however have progenitor properties which can rescue mice from lethal myeloablation; 2) single cell characterisation by combined scRNAseq and scATACseq is not sufficient to identify cells with repopulation capacity; 3) expanded HSCs can be engrafted in unconditioned host and return to quiescence.

The authors also confirm previous studies that EPCR^{high} HSCs have better reconstitution capability than EPCR^{low} HSCs when transplanted.

Strengths:

The major strength of this manuscript is that it describes how functional HSCs are expanded in PVA cultures to a deeper extent than what has been done in the original publication. The authors are also mindful of considering the complexities of interpreting transplantation data. As these PVA cultures become more widely used by the HSC community, this manuscript is valuable as it provides a better understanding of the model and its limitations.

Novelty aspects include:

- The authors determined that small numbers of expanded HSCs enable transplantation into non-conditioned syngeneic recipients.
- This is to my knowledge the first report characterising output of PVA cultures by multiome. This could be a very useful resource to the field.
- They are also the first to my knowledge to use barcoding to quantify HSC repopulation capacity at the clonal level after PVA culture.
- It is also useful to report that HSCs isolated from fetal livers do expand less than their adult counterparts in these PVA cultures.

Weaknesses:

- The analysis of the multiome experiment is limited. The authors do not discuss what cell types, other than functional or phenotypic HSCs are present in these cultures (are they mostly progenitors or bona fide mature cells?) and no quantifications are provided. It seems nonetheless that most cells in these cultures do not acquire differentiation markers. In addition, the functional experiments demonstrate very few retain transplantation capacity. Future works will have to investigate the nature of the bulk of the other cells in these cultures.
- Barcoding experiments are technically elegant but do not bring particularly novel insights.
- Number of mice analysed in certain experiments is fairly low (Figure 1 and 5).
- The manuscript remains largely descriptive. While the data can be used to make useful recommendations to future users working with PVA cultures and in general with HSCs, those recommendations could be more clearly spelled out in the discussion.
- The authors could have provided discussion of the other publications/preprints which have used these methods to date. This would have been useful for researchers who have not used this technique.

Overall, the authors succeeded in providing a useful set of experiments to better interpret what type of HSCs are expanded in PVA cultures. More in depth mining of their bioinformatic data (by the authors or other groups) is likely to highlight other interesting/relevant aspects of HSC biology in relation to this expansion methodology.

<https://doi.org/10.7554/eLife.91826.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Reviewer #1 (Public Review):

In 2019, Wilkinson and colleagues (PMID: 31142833) managed to break the veil in a 20-year open question on how to properly culture and expand Hematopoietic Stem Cells (HSCs). Although this study is revolutionizing the HSC biology field, several questions regarding the mechanisms of expansion remain open. Leveraging on this gap, Zhang et al.; embarked on a much-needed investigation regarding HSC self-renewal in this particular culturing setting.

The authors firstly tackled the known caveat that some HSC membrane markers are altered during in vitro cultures by functionally establishing EPCR (CD201) as a reliable and stable HSC marker (Figure 1), demonstrating that this compartment is also responsible for long-term hematopoietic reconstitution (Figure 3). Next in Figure 2, the authors performed single-cell omics to shed light on the potential mechanisms involved in HSC maintenance, and interestingly it was shown that several hematopoietic populations like monocytes and neutrophils are also present in this culture conditions, which has not been reported. The study goes on to functionally characterize these cultured HSCs (cHSC). The authors elegantly demonstrate using state-of-the-art barcoding strategies that these culturing conditions provoke heterogeneity in the expanding HSC pool (Figure 4). In the last experiment (Figure 5), it was demonstrated that cHSC not only retain their high EPCR expression levels but upon transplantation, these cells remain more quiescent than freshly-isolated controls.

Taken together, this study independently validates that the proposed culturing system works and provides new insights into the mechanisms whereby HSC expansion takes place.

Most of the conclusions of this study are well supported by the present manuscript, some aspects regarding experimental design and especially the data analysis should be clarified and possibly extended.

1. The first major point regards the single-cell (sc) omics performed on whole cultured cells (Figure 2):

a. The authors claim that both RNA and ATAC were performed and indeed some ATAC-seq data is shown in Figure 2B, but this collected data seems to be highly underused.

We appreciate the opportunity to clarify our analytical approach and the rationale behind it. In our study, we employed a novel deep learning framework, SAILERX, for our analysis. This framework is specifically designed to integrate multimodal data, such as RNAseq and ATACseq. The advantage of SAILERX lies in its ability to correct for technical noise inherent in sequencing processes and to align information from different modalities. Unlike methods that force a hard alignment of modalities into a shared latent space, SAILERX allows for a more refined integration. It achieves this by encouraging the local structures of the two modalities, as measured by pairwise similarities.

To put it more simply, SAILERX combines RNAseq and ATACseq data, ensuring that the unique characteristics of each data type are respected and used to enhance the overall biological picture, rather than forcing them into a uniform framework.

While it is indeed possible to analyze the ATAC-seq and RNA-seq modalities separately, and we acknowledge the potential value in such an approach, our primary objective in this study was to highlight the relatively low content of HSCs in cultures. This finding is a key point of our work, and the multiome data support this from a molecular point of view.

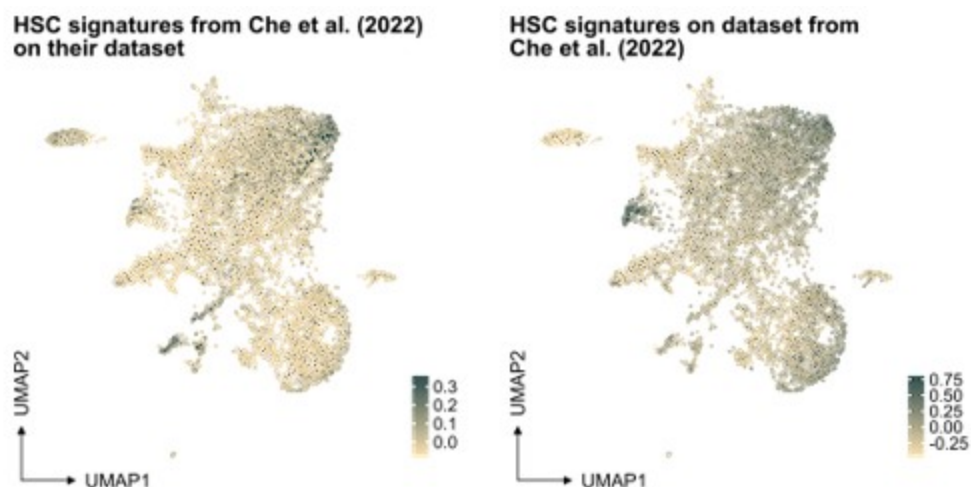
The Seurat object we provide was created to facilitate further analysis by interested researchers. This object simplifies the exploration of both the ATAC-seq and RNA-seq data, allowing for additional investigations that may be of interest to the scientific community. We hope this explanation clarifies our methodology and its implications.

b. It's not entirely clear to this reviewer the nature of the so-called "HSC signatures"(SF2C) and why exactly these genes were selected. There are genes such as Mpl and Angpt1 which are used for Mk-biased HSCs. Maybe relying on other HSC molecular signatures (PMID: 12228721, for example) would not only bring this study more into the current field context but would also have a more favorable analysis outcome. Moreover reclustering based on a different signature can also clarify the emergence of relevant HSC clusters.

In our study, the selection of the HSC signature in our work was based on well-referenced datasets on well-defined HSPCs, as detailed in the "v. HSC signature" section of our methods. This signature was projected also to another single-cell RNA sequencing dataset generated from ex vivo expanded HSC culture (PMID: 35971894, see Author response image 1 below), demonstrating again an association primarily to the most primitive cells (at least based on gene expression).

Author response image 1.

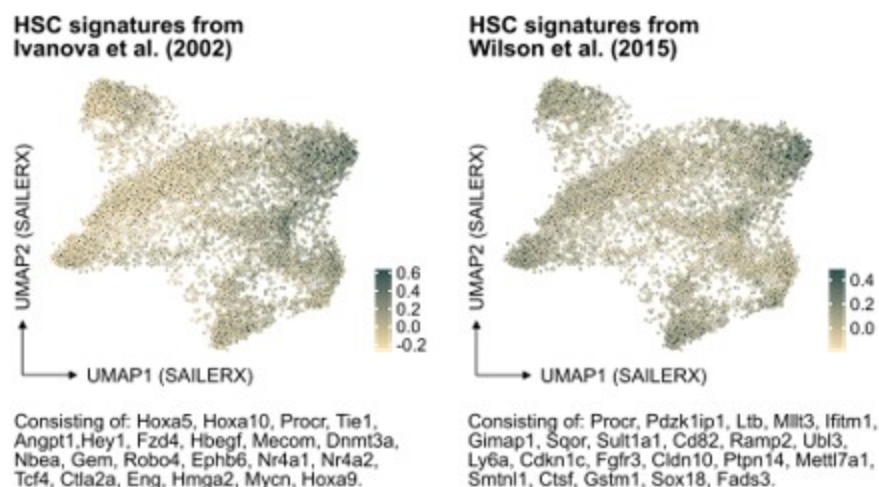
Projection of "our" HSC signature on scRNAseq data from independent work.



In further response to the suggestion here, we have also examined the molecular signature of HSCs referenced in PMID: 12228721 but also of another HSC signature from PMID: 26004780 in our data (Author response image 2). While these signatures do indeed enrich for cells that fall in the cluster of molecularly defined HSCs, our analysis indicates that neither of them significantly improves the identification of HSCs in our dataset compared to the signature we originally used. This finding reinforces our confidence in the appropriateness of our chosen HSC signature for this study.

Author response image 2.

Projection of alternative HSC signatures onto the SAILERX UMAP.



Regarding the specific genes *Mpl* and *Angpt1*, we respectfully oppose the view that these genes are exclusively associated with MK-biased HSCs. There is substantial evidence supporting the broader role of *Mpl* in regulating HSCs, regardless of any particular "lineage bias". Similarly, while *Angpt1* has been less extensively studied, its role in HSCs, as examined in PMID: 25821987, suggests a more general association with HSCs rather than a specific impact on MKs. Therefore, we maintain that it is more accurate to consider these genes as HSC-associated rather than restricted to MK-biased HSCs.

Finally, addressing the comment on reclustering based on different signatures, we would like to clarify that the clustering process is independent of the projection of signatures. The clustering aims to identify cell populations based on their overall molecular profiles, and while signatures can aid in characterizing these populations, they do not influence the clustering process itself.

c. The authors took the hard road to perform experiments with the elegant HSC-specific Fgd5-reporter, and they claim in lines 170-171 that it "failed to clearly demarcate in our single-cell multimodal data". This seems like a rather vague statement and leads to the idea that the scRNA-seq experiment is not reliable. It would be interesting to show a UMAP with this gene expression regardless and also potentially some other HSC markers.

We understand the concerns raised about our statement on the performance of the *Fgd5*-reporter in our multimodal data analysis. Our aim was not to suggest that single-cell molecular data are unreliable. Instead, we intended to point out specific challenges associated with scRNA sequencing, notably the high rates of dropout. Regarding the specific example of *Fgd5*, it appears this transcript is not efficiently captured by 10x technology. Our previous 10x scRNA-seq experiments on cells from the *Fgd5* reporter strain (S  w  n et al., eLife 2018; Konturek-Ciesla et al., Cell Rep. 2023) support this observation. Despite cells being sorted as *Fgd5*-reporter positive, many showed no detectable transcripts.

We consider it pertinent to note that our study integrates ATAC-seq data in conjunction with single-cell molecular data. We believe that this integration, coupled with the analytical methods we have employed, potentially offers a way to address some of the limitations typically associated with scRNA sequencing. However, in assessing frequencies, we observe that the number of candidate HSCs identified via single-cell molecular data is substantially

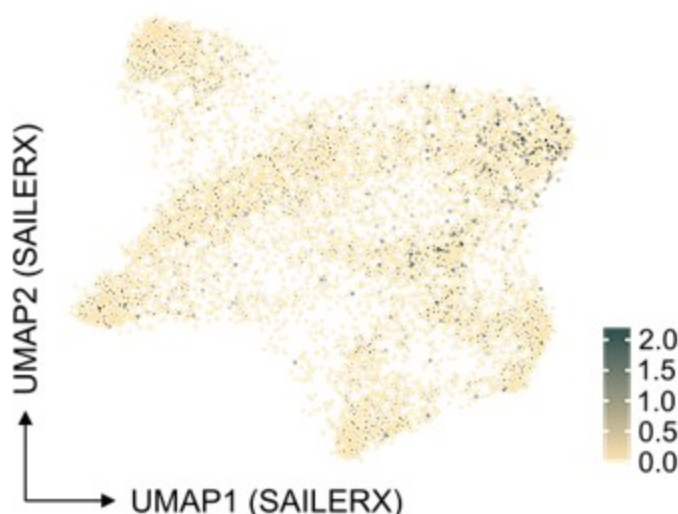
higher compared to those identified through flow cytometry, the latter which we demonstrate correlate functionally with genuine long-term repopulating activity.

With respect to *Fgd5*, as depicted in our analysis below, there appears to be an enrichment of cells in the cluster identified as HSCs, as well as a significant representation in the cycling cell cluster (Author response image 3). Regarding the projection of other individual genes, the Seurat object we have provided allows for such projections to be readily performed. This offers an opportunity for further exploration and validation of our findings by interested researchers.

Author response image 3.

Feature plot depicting *Fgd5* expression in the SAILERX UMAP.

Fgd5 expression on whole culture



1. During the discussion and in Figure 4, the authors ponder and demonstrate that this culturing system can provoke divert HSC close expansion, having also functional consequences. This a known caveat from the original system, but in more recent publications from the original group (PMID: 36809781 and PMID: 37385251) small alterations into the protocol seem to alleviate clone selection. It's intriguing why the authors have not included these parameters at least in some experiments to show reproducibility or why these studies are not mentioned during the discussion section.

Thank you for pointing out the recent publications (PMID: 36809781 and PMID: 37385251) that discuss modifications to the HSC culturing system. We appreciate the opportunity to address why these were not included in our discussion or experiments.

Firstly, it is important to note that these papers were published after the submission of our manuscript. In fact, one of the studies (PMID: 36809781) references the preprint version of our work on Biorxiv. This timing meant that we were unable to consider these studies in our initial manuscript or incorporate any of their findings into our experimental designs.

Furthermore, as strong advocates for the peer-review system, we prioritize references that have undergone this rigorous process. Preprints, while valuable for early dissemination of research findings, do not offer the same level of scrutiny and validation as peer-reviewed publications. Our approach was to rely on the most relevant and rigorously reviewed

literature available to us at the time of submission. This included, most notably, the original and ground-breaking work by Wilkinson et al., which provided a foundational basis for our research.

We acknowledge that the field of HSC research is rapidly evolving, and new findings, such as those mentioned, are continually emerging. These new studies undoubtedly contribute valuable insights into HSC culturing systems and their optimization. However, given the timing of their publication relative to our study, we were not able to include them in our analysis or discussion.

1. In this reviewer's opinion, the finding that transplanted cHSC are more quiescent than freshly isolated controls is the most remarkable aspect of this manuscript. There is a point of concern and an intriguing thought that sprouts from this experiment. It is empirical that for this experiment the same HSC dose is transplanted between both groups. This however is technically difficult since the membrane markers from both groups are different. Although after 8 weeks chimerism levels seem to be the same (SF5D) for both groups, it would strengthen the evidence if the author could demonstrate that the same number of HSCs were transplanted in both groups, likely by limiting dose experiments. Finally, it's interesting that even though EE100 cells underwent multiple replication rounds (adding to their replicative aging), these cells remained more quiescent once they were in an in vivo setting. Since the last author of this manuscript has also expertise in HSC aging, it would be interesting to explore whether these cells have "aged" during the expansion process by assessing whether they display an aged phenotype (myeloid-skewed output in serial transplantations and/or assessing their transcriptional age).

We thank the reviewer for the insightful observations regarding the quiescence of transplanted cultured HSCs. We appreciate the opportunity to clarify the experimental design and its implications, particularly in the context of HSC aging.

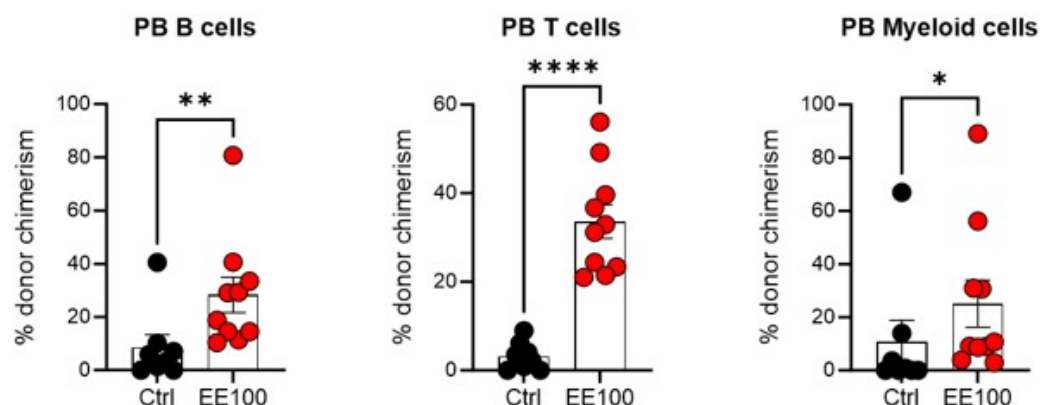
The primary aim of comparing cKit-enriched bone BM cells with cultured cells was to investigate if ex vivo activated HSCs exhibit a similar proliferation pattern to in vivo quiescent HSCs post-transplantation. This comparison was crucial for evaluating the similarity between in vitro cultured and "unmanipulated" HSC behavior. While we acknowledge the technical challenge of transplanting equivalent HSC doses between groups due to differing membrane markers, our study design focused on assessing stem cell activity post-culture. This was quantitatively evaluated by calculating the repopulating units (detailed in Table 1 and Fig S4G), rather than through a limiting dilution assay. There exists a plethora of literature demonstrating the correlation between these assays, although of course the limiting dilution assay is designed to provide a more exact output.

Regarding the intriguing aspect of HSC aging in the context of ex vivo expansion, our observations indicate that both the subfraction of ex vivo expanded cells (Fig 3 and Fig S3) and the entire cultured population (Fig 4B, Fig 5B, Fig S4A, and Fig S5B) maintain long-term multilineage reconstitution capacity post-transplantation. This suggests that the PVA-culture system does not lead to apparent signs of "HSC aging," despite the cells undergoing active self-renewal in vitro. This is further supported by our serial transplantation experiments, where cultured cells continued to demonstrate multilineage capacity rather than any evident myeloid-biased reconstitution 16 weeks post-second transplantation (see Author response image 4 below).

Author response image 4.

Serial transplantation behavior of ex vivo expanded HSCs. 5 million whole BM cells from primary transplantation were transplanted together with 5 million competitor whole BM

cells. The control group was transplanted with 100 cHSCs freshly isolated from BM for the primary transplantation. Mann-Whitney test was applied and the asterisks indicate significant differences. *, $p < 0.05$; **, $p < 0.01$; ****, $p < 0.0001$. Error bars denote SEM.

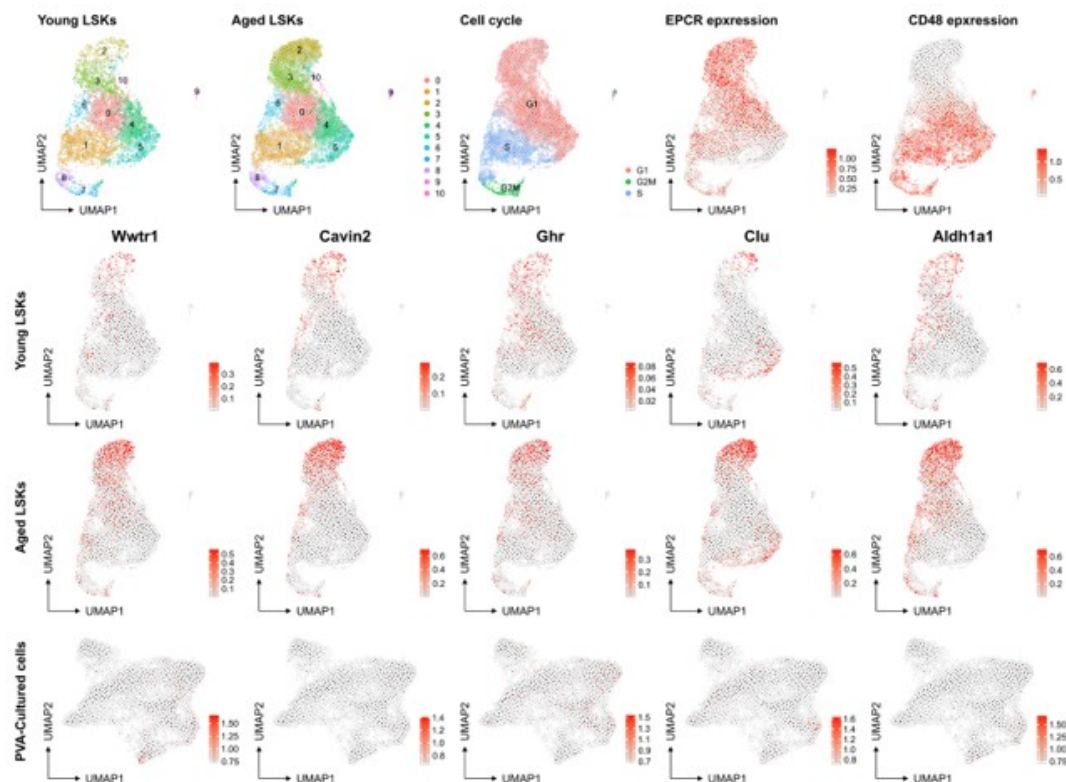


However, we recognize the complexity of defining HSC aging and the potential for the culture system to influence certain aspects of this process. The association of aging signature genes with HSC primitiveness and young signature genes with differentiation presents an interesting dichotomy. Our analysis of a native dataset on young mice and the projection of aged signatures onto our multiome data (as shown below for a set of genes known to be induced at higher levels in aged HSCs (f.i. Wahlestedt et al., Nature Comm 2017), aging scRNAseq data from PMID: 36581635) does not directly indicate that the culture system promotes HSC aging compared to aged Lin-Sca+Kit+ cells. Yet, we do not rule out the possibility that culturing may influence other facets of the HSC aging process.

In conclusion, while our current data do not provide direct evidence of induced HSC aging through the culture system, this remains a compelling area for future research. The potential impact of ex vivo culture on aspects of the HSC aging process warrants further exploration, and we appreciate your suggestion in this regard.

Author response image 5.

No evident signs of "molecular aging" following ex vivo expansion of HSCs. Young and aged scRNAseq data from PMID: 36581635 were integrated and explored from the perspective of known genes associating to HSC aging. The top row depicts contribution to UMAPs from young and aged cells (two left plots), cell cycle scores of the cells, and the expression of EPCR and CD48 as examples markers for primitive and more differentiated cells, respectively. The expression of the HSC aging-associated genes *Wwtr1*, *Cavin2*, *Ghr*, *Clu* and *Aldh1a1* was then assessed in the data as well as in the SAILERX UMAP of cultured HSCs (bottom row).



Reviewer #2 (Public Review):

Summary:

In this study, Zhang and colleagues characterise the behaviour of mouse hematopoietic stem cells when cultured in PVA conditions, a recently published method for HSC expansion (Wilkinson et al., Nature, 2019), using multiome analysis (scRNA-seq and scATACseq in the same single cell) and extensive transplantation experiments. The latter are performed in several settings including barcoding and avoiding recipient conditioning. Collectively the authors identify several interesting properties of these cultures namely: 1) only very few cells within these cultures have long-term repopulation capacity, many others, however, have progenitor properties that can rescue mice from lethal myeloablation; 2) single-cell characterisation by combined scRNAseq and scATACseq is not sufficient to identify cells with repopulation capacity; 3) expanded HSCs can be engrafted in unconditioned host and return to quiescence.

The authors also confirm previous studies that EPCRhigh HSCs have better reconstitution capability than EPCRlow HSCs when transplanted.

Strengths:

The major strength of this manuscript is that it describes how functional HSCs are expanded in PVA cultures to a deeper extent than what has been done in the original publication. The authors are also mindful of considering the complexities of interpreting transplantation data. As these PVA cultures become more widely used by the HSC community, this manuscript is valuable as it provides a better understanding of the model and its limitations.

Novelty aspects include:

- *The authors determined that small numbers of expanded HSCs enable transplantation into non-conditioned syngeneic recipients.*
- *This is to my knowledge the first report characterising the output of PVA cultures by multiome. This could be a very useful resource for the field.*
- *They are also the first to my knowledge to use barcoding to quantify HSC repopulation capacity at the clonal level after PVA culture.*
- *It is also useful to report that HSCs isolated from fetal livers do expand less than their adult counterparts in these PVA cultures.*

Weaknesses:

- *The analysis of the multiome experiment is limited. The authors do not discuss what cell types, other than functional or phenotypic HSCs are present in these cultures (are they mostly progenitors or bona fide mature cells?) and no quantifications are provided.*

The primary objective of our manuscript was to characterize the features of HSCs expanded from ex vivo culture. In this context, our analysis of the single cell multiome sequencing data was predominantly centered on elucidating the heterogeneity of cultures, along with subsequent in vivo functional analysis. This focus is reflected in our comparisons between the molecular features of ex vivo cultured candidate HSCs (cHSCs) and "fresh/unmanipulated" HSCs, as illustrated in Figures 2D-E of our manuscript.

Our findings provide substantial evidence that ex vivo expanded cells share significant similarities with HSCs isolated from the BM in terms of molecular features, differentiation potential, heterogeneity, and in vivo stem cell activity/function. This suggests that the ex vivo culture system closely mimics several aspects of the in vivo environment, thereby broadening the potential applications of this system for HSC research.

Regarding the presence of other cell types in the cultures, it is important to note that most cells did not express mature lineage markers, suggesting their immature status. However, we acknowledge the presence of some mature lineage marker-positive cells within the cultures. These cells are represented by the endpoints in our SAILERX UMAP, indicating a progression from immature to more differentiated states within the culture system.

While the main emphasis of our study was on HSCs, we understand the importance of acknowledging and briefly discussing the presence and characteristics of other cell types in the cultures. This aspect provides a more comprehensive understanding of the culture system and its impact on cellular heterogeneity, although it was for the most part beyond the scope of our studies.

- *Barcoding experiments are technically elegant but do not bring particularly novel insights. We respectfully disagree with the view that our barcoding experiments do not offer novel insights. We believe that the application of barcoding technology in our study represents a significant advancement over previous methods, both in terms of quantitative rigor and ethical considerations.*

In the foundational work by Wilkinson et al., clonal assessments were indeed performed, but these were limited in scope and largely served as proof of concept. Our use of barcoding technology, on the other hand, allowed for a comprehensive quantitative assessment of the expansion potential of HSC clones. This technology enabled us to rigorously quantify the number of HSC clones capable of undergoing at least three self-renewing divisions (e.g. those clones present in 5 separate animals), while also revealing the heterogeneity in their expansion potential.

One alternative approach could have been to culture single HSCs and distribute the progeny among multiple mice for analysis. However, when considering the sheer number of mice that would be required for such an experiment for quantitative assessments, it becomes evident that viral barcoding is a far superior method. Not only does it provide a more efficient and scalable approach to assessing clonal expansion, but it also significantly reduces the number of animals required for the study, aligning with the principles of ethical research and animal welfare.

In conclusion, we assert that the barcoding experiments conducted in our study are not only technically robust but also yield novel quantitative insights into the dynamics of HSC clones within expansion cultures. These insights have value not only for current research but also hold potential implications for future applications.

- *The number of mice analysed in certain experiments is fairly low (Figures 1 and 5).*

We would like to clarify our approach in the context of the 3R (replacement, refinement, and reduction) policy, which guides ethical considerations in animal research.

In alignment with the 3R principles, our study was designed to minimize the use of experimental animals wherever possible. For most experiments, including those presented in Figures 1 and 5, we adopted a standard of using five mice per group. Based on the effect sizes we observed, we concluded that this sample size was appropriate for most parts of our study.

Specifically for Figure 5, we used two animals per time point, totaling seven animals per treatment group. It is important to note that we did not monitor the same animals over time but used different animals at each time point, as mice had to be sacrificed for the type of analyses conducted. Despite the seemingly small sample size, the results we obtained were remarkably consistent across groups. This consistency provided strong evidence that *ex vivo* activated HSCs return to a more quiescent state after being transplanted into unconditioned recipients. Given the clear and consistent nature of these results, we determined that including more animals for the purpose of additional statistical analysis was not necessary.

Our approach reflects a balance between adhering to ethical standards in animal research and ensuring the scientific validity and reliability of our findings. We believe that the sample sizes chosen for our experiments are justified by the consistent and significant results we obtained, which contribute meaningfully to our understanding of HSC behavior post-transplantation.

- *The manuscript remains largely descriptive. While the data can be used to make useful recommendations to future users working with PVA cultures and in general with HSCs, those recommendations could be more clearly spelled out in the discussion.*

We fully agree that many aspects of our study are indeed descriptive, which is reflective of the exploratory and foundational nature of this type of research.

We have strived to provide clear and direct recommendations for researchers interested in utilizing the PVA culture system, which we believe are evident throughout our manuscript:

1. Utility of Viral Delivery in HSC Research: Our research, particularly through the use of barcoding experiments, underscores the effectiveness of viral delivery methods in HSC studies. While barcoding itself is a significant tool, it is the underlying process of viral delivery that truly exemplifies the potential of this approach. Our work shows that the culture system is highly conducive to maintaining HSC activity, which is critical for genetic manipulation. This is evident not only in our current study but also in our previous work that included for transient delivery methods (Eldeeb et al., Cell Reports 2023).
2. Non-conditioned transplantation: Our findings suggest that non-conditioned transplantation can be a valuable method in studying both normal and malignant hematopoiesis. This approach can complement genetic lineage tracing models, providing a more native and physiological context for hematopoietic research. We state this explicitly in our discussion.
3. Integration with recent technical advances: The combination of the PVA culture system with recent developments in transplantation biology, genome engineering, and single-cell technologies holds significant promise. This integration is likely to yield exciting discoveries with relevance to both basic and clinically oriented hematopoietic research. This is the end statement of our discussion.

While our manuscript is in a way tailored to those with experience in HSC research, we have made a concerted effort to ensure that the content is accessible and informative to a broader audience, including those less familiar with this area of study. Our intention is to provide a resource that is both informative for experts in the field and approachable for newcomers.

• *The authors should also provide a discussion of the other publications that have used these methods to date.*

We would like to clarify that the scope of literature on the specific methods we employed, particularly in the context of our research objectives, is not extensive. Most of the existing references on these methods come from a relatively narrow range of research groups. In preparing our manuscript, we tried to be comprehensive yet selective in our citations to maintain focus and relevance. Our referencing strategy was guided by the aim to include literature that was most directly pertinent to our study's methodologies and findings.

Overall, the authors succeeded in providing a useful set of experiments to better interpret what type of HSCs are expanded in PVA cultures. More in-depth mining of their bioinformatic data (by the authors or other groups) is likely to highlight other interesting/relevant aspects of HSC biology in relation to this expansion methodology.

We are grateful for the overall positive assessment of our work and the recognition of its contributions to understanding HSC expansion in PVA cultures.

We agree that every study, including ours, has its limitations, particularly regarding the scope and depth of exploration. It is challenging to cover every aspect comprehensively in a single study. Our research aimed to provide a foundational understanding of HSCs in PVA cultures, and we are pleased that this goal appears to have been met.

We also concur with your point on the potential for further in-depth mining of our bioinformatic data. Our hope is that this data can serve as a resource (or at least a starting point) for other investigators.

In conclusion, we hope that our responses have adequately addressed your queries and clarified any concerns. We are committed to contributing to the growth of knowledge in HSC

research and look forward to the advancements that our study might enable, both within our team and the wider scientific community.

Reviewer #1 (Recommendations For The Authors):

1. In Line 150, the R packages can/should be mentioned just in the method section;

We have moved this text to the methods section.

1. In Figure F3C adding a legend next to the plot would assist the reader in identifying which populations are referred to, as the same color pellet is used for other panels;

We have now adjusted the figure legend position to make it more clear for the reader.

1. In Figure 4D, for the pre-culture experiments 1000 cHSCs were used and then in the post-culture 1200 cHSCs were used. Can the authors justify the different numbers?

The decision to use 1000 cHSCs in the pre-culture experiments and 1200 cHSCs in the post-culture experiments was not based on a specific rationale favoring one cell number over the other. In our Method section, we have detailed our experimental design, which was structured to provide robust and reliable readouts of HSC behavior and characteristics in different conditions.

We consider the two cell numbers – 1000 and 1200 – to be quite similar in the context of our experimental aims. Since the readouts here are based on clonal assessments, this slight difference in cell numbers is unlikely to significantly impact the overall conclusions drawn from these experiments. The primary focus of our study was on qualitative aspects of HSC behavior and function, rather than on quantitative differences that might arise from small variations in initial cell numbers.

1. In SF5F it would help readers if a line plot (per group) was also shown together with the dot plots. Moreover, applying statistics to the trend lines (Wilcoxon, for example) would strengthen the argument that cHSCs divide less than control cells.

We would like to clarify that the data presented in SF5F were derived from different animals at each respective time point. As such, the data points at each time point represent independent measurements from separate animals, rather than a continuous measurement from the same set of animals over time. Therefore, creating a line plot that connects each time point within a group would inadvertently convey a misleading impression of a longitudinal study on the same animals, which is not reflective of the actual experimental design. Instead, the dot plot format was chosen as it more accurately depicts the independent and discrete nature of the measurements at each time point. Our current data presentation method was selected to provide the most accurate and transparent representation of our findings.

Reviewer #2 (Recommendations For The Authors):

Listed below are recommendations to further improve this manuscript:

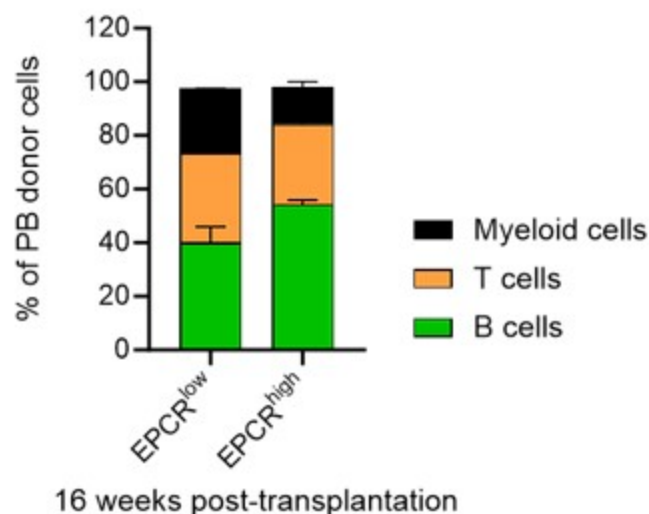
Major Comments

1. Fig 1: the authors showed that EPCR^{high} HSCs have better reconstitution capability than EPCR^{low} HSCs via bone marrow transplantation. Additionally, mice receiving cultured EPCR^{high} SLAM LSK cells were more efficiently radioprotected than those receiving PVA expanded EPCR^{low} SLAM LSK.

a. In addition to Fig.1F, authors should show the lineage distributions and chimerism of mice receiving cultured EPCR^{high} and EPCR^{low} SLAM LSK respectively.

We have indeed analyzed the lineage distribution in these experiments, and our findings indicate no statistically significant differences between the groups (see graph in Author response image 6). This suggests that the cultured EPCR^{high} and EPCR^{low} SLAM LSK cells do not preferentially differentiate into specific lineages in a way that would impact the overall interpretation of our results.

Author response image 6.



Regarding the chimerism in peripheral blood (PB) lineages, Fig. 1F in our manuscript currently shows the PB myeloid chimerism. We chose to focus on this parameter as it most directly relates to our study's objectives. We did here not transplant with competitor cells, and in most cases, the chimerism levels reached 100% for lineages other than T cells (T cells being more radioresistant). Based on our analysis, including data on chimerism in other PB lineages would not significantly enhance the understanding of the functional capacity of the transplanted cells, as the myeloid chimerism data already provides a robust indicator of their engraftment and functional potential.

We believe that our current presentation of data in Fig. 1F, along with the additional analyses provided in the results section, offers a comprehensive understanding of the behavior and potential of the cultured EPCR^{high} and EPCR^{low} SLAM LSK cells.

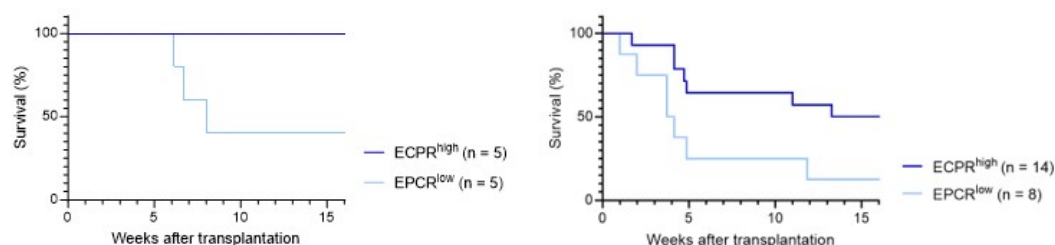
b. Fig1F: only 5 mice were used in each group. Could this result occur by chance? Testing with Fisher's exact test with the data provided results in $p=0.16$. The authors should consider adding more animals or adding the p -value above (or from another relevant test) for readers' consideration.

We acknowledge the point that only five mice were used in each group and understand the concern regarding the robustness of our findings.

As correctly noted, applying Fisher's exact test to the data in Fig. 1F results in a p-value which does not reach the conventional threshold for statistical significance. However, one might also consider the analysis of the KM survival curve, which associated with a p-value of 0.0528 (Fig. 1F, left graph below; Gehan-Breslow-Wilcoxon test). A similar test on the single-cell culture transplantation experiment (Fig. 1E, right graph below) also demonstrated statistical significance (p-value = 0.0485).

While these p-values meet (or are very close to) the conventional criteria for statistical significance ($p < 0.05$), we have chosen to place greater emphasis on effect sizes rather than strictly on p-values. This decision is based on our belief that effect sizes provide a more direct and meaningful measure of the biological impact observed in our experiments. We find that the effect sizes observed are compelling and consistent with the overall narrative of our study.

Author response image 5.



1. The characterisation of the multiome experiment is highly underdeveloped.

a. From an experimental point of view, it is not clear how the PVA culture for this experiment was started. Are there technical/biological replicates? Have several PVA cultures been pooled together?

We have included these details in the revised text to ensure a comprehensive understanding of our experimental setup.

b. Fig2B: The authors should present more data as to how each of the clusters was annotated (bubble plot of marker genes used for annotation?) and importantly the percentage of cells in each of the clusters. It is particularly relevant to note what % is the cluster annotated as HSCs and compare that to the % of phenotypic HSCs and the % repopulating HSCs calculated in the transplantation experiments.

In our study, the annotation of clusters was primarily based on reference genes for cell types from prior works in the field, such as from our recent work (Konturek-Ciesla et al., Cell Reports 2023). Additionally, we employed transcription factor (TF) motifs to assign identities to these clusters. This approach is relatively standard in the field, and we believe it provides a robust framework for our analysis. We included information on some of the key TF motifs used to guide our annotations.

Regarding the assignment of a percentage to cells within the HSC cluster, we initially had reservations about the utility of this measure. This is because the transcriptional identity of HSCs might not align precisely with their identity based on candidate HSC protein markers. There are complexities related to transcriptional continuums that could influence the interpretation of such data. However, acknowledging your request for this information, we have now included the percentage of cells in the HSC cluster in Fig. 2B for reference.

We also wish to highlight that when isolating EPCR+ cells, which encompasses a range of CD48 expression, clustering becomes much less distinct, as shown in Fig. 2E. Most of these cells do not demonstrate long-term functional HSC activity in a transplantation setting (as presented in Figure 3). This observation underscores the challenges in deducing HSC identity based solely on molecular data and reinforces the importance of functional validation.

c. Are there any mature cells in these PVA cultures? The annotations presented in the table under the UMAP are vague: Are cluster 4 monocytes or monocytes progenitors? Same for clusters 0,1 and 7 - are these progenitors or more mature cells? How were HPCs (cluster 3) distinguished from cHSCs (cluster 5)?

We agree with your observation that the annotations for certain clusters, such as clusters 4, 0, 1, and 7, as well as the distinction between HPCs (cluster 3) and cHSCs (cluster 5), appear vague. This vagueness to some extent stems from the challenges inherent in comparing cultured cells to their counterparts isolated directly from animals. Most reference data defining cell types are derived from cells in their native state, and less is known about how these definitions translate to the progeny of HSPCs cultured in vitro.

In our study, we used the expression of reference genes and enriched transcription factor motifs to annotate clusters. This method, while useful, has its limitations in precisely defining the maturation stage of cells in culture. The enrichment of lineage-defining factors at the ends of the UMAP suggests the presence of more mature cells, whereas the lack of lineage marker expression in the majority of cells implies a general lack of terminal differentiation.

This issue is not necessarily unique to the culture situation, as similar challenges in cell type annotation are encountered in other contexts, such as the analysis of granulocyte-macrophage progenitors in bone marrow, where a vast range of cell types and clusters are identified (e.g., PMID: 26627738). To try to address these challenges, we employed an approach detailed in the methods section under the header "iv. ATAC processing and cluster annotation." We assessed marker genes for clusters using Enrichr for cell types, relying on databases designed to provide gene expression identities to defined cell types. This methodology informed our references to the clusters.

In summary, while our annotations provide a general overview of the cell types present in the cultures, we acknowledge the complexities and limitations in precisely defining these types, particularly in distinguishing between progenitors and more mature cells. We hope this explanation clarifies our approach and the considerations behind our cluster annotations, but at the same time feel that the alternative approaches have their own drawbacks.

d. What is the meaning of the trajectories presented in Figure 2C? In the absence of a comparison to i) what is observed either when HSCs are cultured in control/non-expanding conditions ii) an in vivo landscape of differentiation in mouse bone marrow; this analysis does not bring any relevant piece of information.

We understand the perspective on comparisons to control conditions and in vivo differentiation landscapes. However, we respectfully disagree with the viewpoint that the analysis that we have performed does not bring relevant information.

The trajectory analysis in Figure 2C is intended to provide insights into the cell types generated in our PVA cultures and the potential differentiation pathways they may follow. This kind of analysis is particularly valuable in the context of understanding how in vitro cultures can support HSC maintenance and differentiation, which is a topic of significant interest in the field. For instance, studies like PMID: 31974159 have highlighted the importance of combining in vitro HSC cultures with molecular investigations.

While we acknowledge that our analysis would benefit from a direct comparison to control or non-expanding conditions, as well as to an in vivo differentiation landscape, we believe that the information provided by our current analysis still holds substantial value. It offers a glimpse into the possible cellular dynamics and differentiation routes within our culture system, which can be a valuable reference point for other investigators working with similar systems.

Regarding the confidence in computed differentiation trajectories, we recognize that this is an area where caution is warranted. Computational approaches to define cell differentiation pathways have inherent limitations and should be interpreted within the context of their assumptions and the data available. This challenge is not unique to our work but is a broader issue in the field of computational biology.

In conclusion, while we agree that additional comparative analyses could further enrich our findings, we maintain that the trajectory analysis presented in Figure 2C contributes meaningful insights into cell differentiation in our PVA culture system. We believe these insights are of interest and value to researchers exploring the complex interplay of HSC maintenance and differentiation in vitro.

1. The addition of barcoding experiments is appreciated. However, it is already known that upon transplantation clonal output is highly heterogeneous, with a small number of clones predominating over others. This is particularly the case after myeloablation conditioning.

a. The "pre-culture" experimental design makes sense. The "post-culture" one is however ambiguous in terms of result interpretation. The authors observe fewer clones contributing to a large proportion of the graft (>5%) than in the "pre-culture" setting. Their interpretation is that expanded HSCs are functionally more homogeneous than the input HSCs. However, in the pre-culture experiment, there are 19 days of expansion during which there will be selection pressures over culture plus ongoing differentiation. In the post-culture experiment, there is no time for such pressures to be exerted. Therefore the conclusion drawn by the authors is not the only conclusion. I would encourage the authors to compare the "pre-culture" experiment to an experiment in which cHSCs are in culture for 48h, then barcoded, and then transplanted. This would be much more informative and would allow a proper comparison of expanded HSCs vs input HSCs.

We understand the perspective that a shorter culture period would reduce the influence of selection pressures and differentiation, potentially allowing for a more direct comparison between expanded HSCs and input HSCs. However, we would like to point out that similar experiments have been conducted in the past, as referenced in our work (PMID: 28224997) and others (PMID: 21964413). These studies have demonstrated a significant heterogeneity in the reconstituting clones when barcoding is done early and cells are transplanted directly.

In light of previous research, we are confident that our methodology — tracking the fates of candidate HSC clones throughout the culture period and assessing the outcomes of individual cells from these expanding clones — yields significant and pertinent insights. We want to highlight the significance of barcoding cells late in the culture, a strategy that allows us to barcode cells that have already been subjected to potential selection pressures within the culture environment. Our primary objective is to investigate the effects of these selection pressures on the subsequent in vivo behavior of the cells that emerge from this process. By focusing on this aspect, we aim to deepen the understanding of how in vitro culture conditions influence the functional characteristics and heterogeneity of HSCs after expansion. We believe this approach provides a unique perspective on the adaptive changes

HSCs undergo during culture and their implications for transplantation efficacy and HSC biology. Our study thus addresses a critical question in the field: how do the conditions and selection pressures inherent to in vitro culture impact the quality and behavior of HSCs upon their return to an in vivo environment?

b. Another experiment the authors may consider is barcoding in unconditioned recipients as there the bottleneck of selecting specific clones should be lower. In addition, this could nicely complement the return to quiescence observed in Figure 5 (see point below)

We agree that this experiment could provide valuable insights, particularly in understanding how different selection pressures might affect HSC clones in various transplantation contexts. It would indeed be a worthwhile complement to our observations in Figure 5 regarding the return to quiescence of HSCs post-transplantation.

However, we would like to point out that our study already includes a substantial amount of data and analyses aimed at addressing specific research questions within this defined scope. The addition of an experiment with barcoding in unconditioned recipients, while undoubtedly relevant and interesting, would extend beyond the boundaries we set for this particular study.

1. Figure 5D-F, only 2 animals per condition were tested, so the experiment is underpowered for any statistics. How about cell viability of cHSC after in vitro culture? The authors have also not tested whether there is a difference in cell viability post-transplant between EE100 and control. In addition, comparing cell cycle profiles of donor EPCR+ HSCs in these transplanted mice would provide additional evidence to support the conclusion.

Regarding the sample size, we acknowledge that only two animals per condition were used in these experiments, which limits the statistical power for robust quantitative analysis. This decision was guided by ethical considerations to minimize animal use, in line with the 3Rs principle (Replacement, Reduction, Refinement). Despite the small sample size, we believe that the strong trends observed in these experiments are indicative and consistent with our broader findings, although we recognize the limitations in terms of statistical generalization. At the same time, as we have written in the public response: "Specifically for Figure 5, we used two animals per time point, totaling seven animals per treatment group. It is important to note that we did not monitor the same animals over time but used different animals at each time point, as mice had to be sacrificed for the type of analyses conducted."

In the context of post-transplant analysis, conducting separate viability assessments on transplanted cells is not typically informative. This is because non-viable cells would naturally be eliminated through biological processes such as phagocytosis soon after transplantation. Therefore, any post-transplant viability analysis would not provide meaningful insights into the engraftment potential or behavior of the transplanted cells.

However, it is important to note that in all our cell isolation and analysis protocols, we routinely include viability markers. This practice ensures that the cell populations we study and report on are indeed viable. Including these markers is a standard part of our methodology and contributes to the accuracy and reliability of our data.

Regarding the comparison of cell cycle profiles, we chose to focus on the cell trace assay as a means to monitor and track cell division history, which directly addresses the central theme here - informing on the proliferation and quiescence dynamics of transplanted HSCs. While comparing cell cycle profiles could perhaps offer an additional layer of information, we did not deem it essential for our core objectives.

1. Several publications have used these PVA cultures and made comments on their strengths and limitations. They do not overlap with this study but should be discussed here for completeness (for example Che et al, Cell Reports, 2022; Becker et al., Cell Stem Cell, 2023; Igarashi, Blood Advances, 2023).

See comments to reviewer 1.

Minor Comments

Figure 1C: should add in the legend that this is in peripheral blood.

Figure 2C: typo in the title.

Figure 3A: typo in "equivalent". We thank the reviewer for catching these errors, which we have now corrected.

Figure 3B and 3C: symbol colours of EPCR^{high}CD48⁺ and EPCR⁻ are too similar to distinguish the 2 groups easily. We highly recommend using contrasting colours.

For easier visualization, we have changed the symbol types and colors in our revised version.

Fig3B and S3A-B: authors should show statistical significance in comparing the 4 fractions. We have now added this information.

In the discussion, the authors rightly point out a paper that described EPCR⁺ HSCs. There are other papers that also looked at EPCR intensity (high vs low), for example, Umemoto et al., EMBO J, 2022.

While we acknowledge the relevance of the paper you mentioned, we faced constraints in the number of references we could include. Therefore, we prioritized citing the original demonstration of EPCR as an HSC marker, particularly focusing on the work by the Mulligan laboratory, which established that cells expressing the highest levels of EPCR exhibit the most potent HSC activity. We believe this reference most directly supports the core focus of our study and provides the necessary context for our findings.